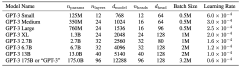
Advanced Artificial Intelligence and Machine Learning

Large Language Models



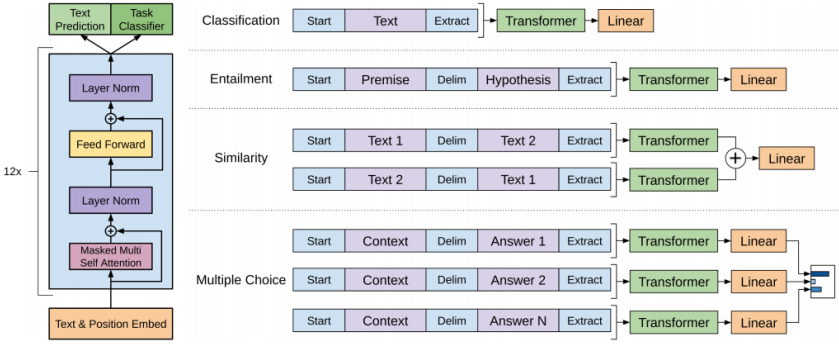
Model Name	$n_{parameters}$	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

Learning goals

- GPT-1 to GPT-3: generative pre-training, zero-/few-shot learning, capabilities and limitations
- Decoding: greedy and beam search, sampling strategies, decoding hyperparameters, evaluating generated text
- Fine-tuning and prompting: instruction tuning, multitask prompted training
- Chain-of-thought prompting and emergent abilities
- Reasoning models: DeepSeek-R1

GPT-1 (2018)

GPT-1



▸ Source: Radford et al., 2018

ARCHITECTURAL DETAILS

- Transformer *decoder* as backbone of the architecture
  - 12-layer-decoder with masked self-attention mechanism
  - Hidden dimension $H = 768$, $A = 12$ Attention heads
  - BPE vocabulary w/ 40k merges
  - Learned positional embeddings (as opposed to fixed, sinusoidal ones in the original Transformer)
- With $U = (w_{t-k}, \ldots, w_{t-1})$

$$\vec{h}_0 = \vec{U}\vec{W}_e + \vec{W}_p$$
$$\vec{h}_l = \mathit{Trafo}(\vec{h}_{l-1}) \forall l \in [1, n]$$
$$P(w_t) = \mathit{softmax}(\vec{h}_n \vec{W}_e^\top)$$

PRE-TRAINING GPT

- Standard LM objective

$$L_1(\{w_1, \ldots, w_n\}) = \sum_i \log(P(w_t | w_{t-k}, \ldots, w_{t-1}; \Theta))$$

  where $\{w_1, \ldots, w_n\}$ is an *unlabeled* sequence of tokens

- *Resource:* BooksCorpus
  - $>$ 7k unpublished books from various genres
  - contains long texts and thus allows learning long range dependencies

FINE-TUNING GPT

- Linear output layer with softmax activation on top
- Auxiliary language modeling objective during fine-tuning
  $\rightarrow$ Improves generalization
  $\rightarrow$ Accelerates convergence
- Task-specific input transformations
  - *Entailment:*
    Concatenation of premise (p) & hypothesis (h): $[p; \$; h]$
  - *Similarity:* Use both orderings and concatenate resulting representations: $[s_1; \$; s_2]$ and $[s_2; \$; s_1]$
  - *Q&A and Commensense Reasoning:*
    Concatenate context (z), question (q) and each possible answer (a_k): $[z; q; \$, a_k]$
- Fine-tuning is rather quick, 3 epochs were sufficient

FINE-TUNING GPT

- Additional objective:

$$L_2(\{w_1, \ldots, w_n\}) = \sum_{x,y} \log(P(y|w_1, \ldots, w_n))$$

  where
  - $P(y|w_1, \ldots, w_n) = softmax(h_l^m W_y)$ and
  - $\{w_1, \ldots, w_n\}$ is a *labeled* sequence of tokens
- Combining both objectives:

$$L_3(\{w_1, \ldots, w_n\}) = L_2(\{w_1, \ldots, w_n\}) + \lambda \cdot L_1(\{w_1, \ldots, w_n\})$$

SOTA RESULTS

Performance on different benchmarks:

Table 2: Experimental results on natural language inference tasks, comparing our model with current state-of-the-art methods. 5x indicates an ensemble of 5 models. All datasets use accuracy as the evaluation metric.

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	89.3	-	-	-
CAFE [58] (5x)	80.2	79.0	89.3	-	-	-
Stochastic Answer Network [35] (3x)	80.6	80.1	-	-	-	-
CAFE [58]	78.7	77.9	88.5	83.3		
GenSen [64]	71.4	71.3	-	-	82.3	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	**61.7**
Finetuned Transformer LM (ours)	**82.1**	**81.4**	**89.9**	**88.3**	**88.1**	56.0

Table 3: Results on question answering and commonsense reasoning, comparing our model with current state-of-the-art methods.. 9x means an ensemble of 9 models.

Method	Story Cloze	RACE-m	RACE-h	RACE
val-LS-skip [55]	76.5	-	-	-
Hidden Coherence Model [7]	77.6	-	-	-
Dynamic Fusion Net [67] (9x)	-	55.6	49.4	51.2
BiAttention MRU [59] (9x)	-	60.2	50.3	53.3
Finetuned Transformer LM (ours)	**86.5**	**62.9**	**57.4**	**59.0**

▸ Source: Radford et al., 2018

GPT PARAMETER COUNT

- **We know**:

$$n_{layers} = 12; \quad d_{model} = 768; \quad V = 40000; \quad M = 512;$$

- **Also**:

$$N_{Decoder} = 12 \cdot d_{model}^2 \quad \textit{and} \quad N_{Embedding} = \overbrace{V \times d_{model}}^{\text{token embedding}} + \underbrace{M \times d_{model}}_{\text{pos. embedding}}$$

$$\begin{aligned} \Rightarrow \quad N_{total} &= n_{layers} \cdot N_{Decoder} + N_{Embedding} \\ &= 12 \cdot 12 \cdot 768^2 + 40000 \times 768 + 512 \times 768 \\ &= 116,047,872 \approx 117M \end{aligned}$$

Note that $N_{Decoder} = 12 \cdot d_{model}^2$ and not $16 \cdot d_{model}^2$ because the Decoder here doesn't do cross attention!

GPT-2 (2019)

STARTING WITH A CONTROVERSY

Release Strategy

Due to concerns about large language models being used to generate deceptive, biased, or abusive language at scale, we are only releasing a much smaller version of GPT-2 along with sampling code. We are not releasing the dataset, training code, or GPT-2 model weights. Nearly a year ago we wrote in the OpenAI Charter: "we expect that safety and security concerns will reduce our traditional publishing in the future, while increasing the importance of sharing safety, policy, and standards research," and we see this current work as potentially representing the early beginnings of such concerns, which we expect may grow over time. This decision, as well as our discussion of it, is an experiment: while we are not sure that it is the right decision today, we believe that the AI community will eventually need to tackle the issue of publication norms in a thoughtful way in certain research areas. Other disciplines such as biotechnology and cybersecurity have long had active debates about responsible publication in cases with clear misuse potential, and we hope that our experiment will serve as a case study for more nuanced discussions of model and code release decisions in the AI community.

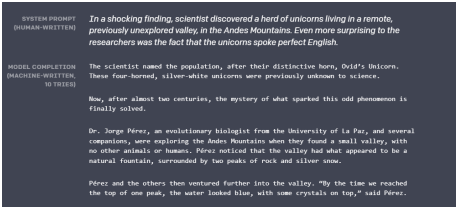
▸ Source: OpenAI Blog

CAPABILITIES – STORYTELLING

SYSTEM PROMPT (HUMAN-WRITTEN)	*In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.*
MODEL COMPLETION (MACHINE-WRITTEN, 10 TRIES)	The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science. Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved. Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow. Pérez and the others then ventured further into the valley. "By the time we reached the top of one peak, the water looked blue, with some crystals on top," said Pérez.

▸ Source: OpenAI Blog

CAPABILITIES – STORYTELLING



▸ Source: OpenAI Blog

- **In 2019:** Great achievement as the model is able to continue a made up story *in a coherent way* by making up its own facts

  Sill: Not very consistent ("`four-horned [...] unicorns`")
- **In 2023:** Nowadays this phenomenon is known as **hallucination** and a lot of research effort is put into mitigating this behaviour

THE ARCHITECTURE

- Transformer decoder pre-trained on AR language modeling
- Custom web scrape (not publicly available) of all outbound links from Reddit

→ 8M documents / 40GB of text

- Byte-level BPE for tokenization
- Increased context size: 512 → 1024

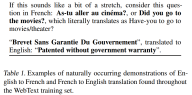
Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

▸ Source: Radford et al., 2019

LANGUAGE MODELS AS MULTITASK LEARNERS

"But as exemplified in McCann et al. (2018), language provides a flexible way to specify tasks, inputs, and outputs all as a sequence of symbols.
For example, a translation training example can be written as the sequence (translate to french, english text, french text)."

- Model benefits from seeing "natural" occurrences of task demonstrations during pre-training on large-scale web corpora
  $\rightarrow$ Related rationale to T5 (*literal task descriptions*), but slightly different (*acquired on the fly already during pre-training*).



If this sounds like a bit of a stretch, consider this question in French: **A-t-u allé au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

"**Brevet Sans Garantie Du Gouvernement**", translated to English: "**Patented without government warranty**".

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

▸ Source: Radford et al., 2019

CAPABILITIES – ZERO-SHOT

- *Zero-shot* refers to solving a task which the model was not previously explicitly trained on, just providing it with a description of the task but no demonstrations, i.e. input/output pairs.
- *Few-shot* would be a relaxation of this setting, allowing for also providing the models with demonstrations (but still no training/gradient updates).
- Radford et al. show GPT-2's zero-shot capabilities on a range of different tasks, paving the way for the developments that came with GPT-3.

CAPABILITIES – ZERO-SHOT

DATASET	METRIC	RESULT	RECORD	HUMAN
Winograd Schema Challenge	accuracy (+)	**70.70%**	63.7%	92%+
LAMBADA	accuracy (+)	**63.24%**	59.23%	95%+
LAMBADA	perplexity (–)	**8.6**	99	~1-2
Children's Book Test Common Nouns (validation accuracy)	accuracy (+)	**93.30%**	85.7%	96%
Children's Book Test Named Entities (validation accuracy)	accuracy (+)	**89.05%**	82.3%	92%
Penn Tree Bank	perplexity (–)	**35.76**	46.54	unknown
WikiText-2	perplexity (–)	**18.34**	39.14	unknown
enwik8	bits per character (–)	**0.93**	0.99	unknown
text8	bits per character (–)	**0.98**	1.08	unknown
WikiText-103	perplexity (–)	**17.48**	18.3	unknown

▸ Source: Radford et al., 2019

(Note: The numbers correspond to the last row of the Table on slide 6)

CAPABILITIES – FACTUAL KNOWLEDGE

Natural Questions ▸ Kwiatkowski et al., 2019

- Testing the *factual knowledge* that is present in the pre-trained model
- Smallest model (117M): $< 1\%$
- GPT-2 (1542M): 4.1%

  $\rightarrow$ Model capacity as a major factor
- Calibration: High accuracy (63.1%) on the 1% most confident answers

CAPABILITIES – FACTUAL KNOWLEDGE

The 30 most confident answers:

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%
Green algae is an example of which type of reproduction?	parthenogenesis	✗	56.5%
Vikram samvat calender is official in which country?	India	✓	55.6%
Who is mostly responsible for writing the declaration of independence?	Thomas Jefferson	✓	53.3%
What us state forms the western boundary of montana?	Montana	✗	52.3%
Who plays ser davos in game of thrones?	Peter Dinklage	✗	52.1%
Who appoints the chair of the federal reserve system?	Janet Yellen	✗	51.5%
State the process that divides one nucleus into two genetically identical nuclei?	mitosis	✓	50.7%
Who won the most mvp awards in the nba?	Michael Jordan	✗	50.2%
What river is associated with the city of rome?	the Tiber	✓	48.6%
Who is the first president to be impeached?	Andrew Johnson	✓	48.3%
Who is the head of the department of homeland security 2017?	John Kelly	✓	47.0%
What is the name given to the common currency to the european union?	Euro	✓	46.8%
What was the emperor name in star wars?	Palpatine	✓	46.5%
Do you have to have a gun permit to shoot at a range?	No	✓	46.4%
Who proposed evolution in 1859 as the basis of biological development?	Charles Darwin	✓	45.7%
Nuclear power plant that blew up in russia?	Chernobyl	✓	45.7%
Who played john connor in the original terminator?	Arnold Schwarzenegger	✗	45.2%

Table 5. The 30 most confident answers generated by GPT-2 on the development set of Natural Questions sorted by their probability according to GPT-2. None of these questions appear in WebText according to the procedure described in Section 4.

▸ Source: Radford et al., 2019

IN THE MEANTIME

GPT-2 Interim Update, May 2019

We're implementing two mechanisms to responsibly publish GPT-2 and hopefully future releases: staged release and partnership-based sharing. We're now releasing a larger 345M version of GPT-2 as a next step in staged release, and are sharing the 762M and 1.5B versions with partners in the AI and security communities who are working to improve societal preparedness for large language models.

Staged Release

Staged release involves the gradual release of a family of models over time. The purpose of our staged release of GPT-2 is to give people time to assess the properties of these models, discuss their societal implications, and evaluate the impacts of release after each stage.

As the next step in our staged release strategy, we are releasing the 345M parameter version of GPT-2. This model features improved performance relative to the 117M version, though falls short of the 1.5B version with respect to the ease of generating coherent text. We have been excited to see so many positive uses of GPT-2-117M, and hope that 345M will yield still more benefits.

▸ Source: OpenAI Blog

IN THE MEANTIME

- Available on huggingface: `https://huggingface.co/gpt2`
- GPT-3 built on the foundation laid by GPT-2
- (also ChatGPT happened)
- GPT-2 (multi-task learning) extended to GPT-3 (prompting)
- Few-/Zero-Shot capabilites of models have become more important (cf. next chapter)
- Models of over 200$\times$ the size of GPT-2 have been trained
- Transformer still the backbone of (nearly) all of them

GPT-3 (2020) & X-shot Learning

GPT RECAP

- Like BERT, GPTs are called "language models".
- Like BERT, GPTs are trained on huge corpora, actually on even huger (closed-source) corpora.
- Like BERT, GPTs are based on the transformer architecture.
- Unlike GPTs, BERT is *not* a language model in the conventional sense, i.e. not an ARLM
- BERT instead relies on the cloze taks, i.e. it is an MLM
- GPTs are conventional ARLMs: they are just trained on predicting the next word (or subword).

GPT RECAP

- Difference 1: GPTs rely on the transformer decoder

  $\rightarrow$ They are called "*generative*" Large Language Models (LLM)

  $\rightarrow$ BERT relies on the encoder, hence not perfectly suited for generation
- Difference 2: GPT is a single model that aims to solve all tasks.
  - It can also switch back and forth between tasks and solve tasks within tasks, another human capability that is important in practice. "fluidity"
- Difference 3: GPT leverages task descriptions.
- Difference 4: GPT is effective at few-shot learning.

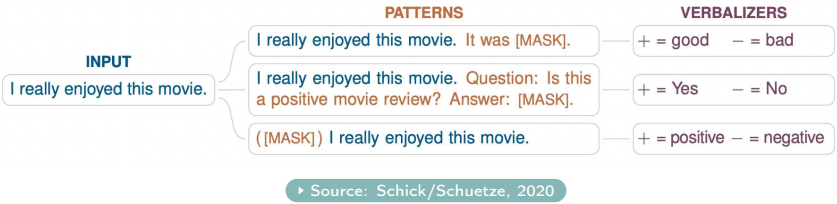
PROMPTING

- Salesforce/GPT2: solve tasks by generation through minor reformatting of original tasks (e.g., generate answer in QA, generate label in classification)
- Prompting: solve tasks by generation through carefully designed prompts/instructions that are very different from the original talk description.
- Prompting is usually used within in-context learning (zero-shot, one-shot, few-shot)
- prompting = instruction = task description = pattern

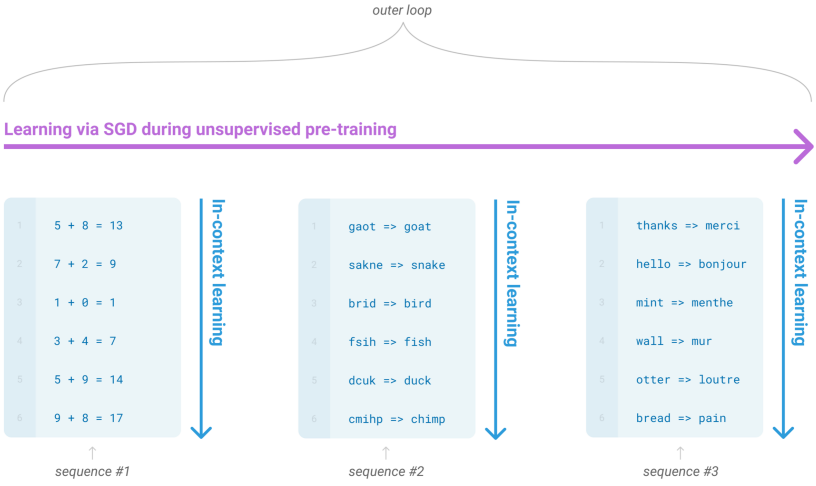
SCHICK (JANUARY 2020)

- Arguably the first proposal of prompting with language models
- GPT3 published several months later
- Schick & Schuetze (2020): Few-Shot Text Generation with Natural Language Instructions (arxiv)

SCHICK (2020)

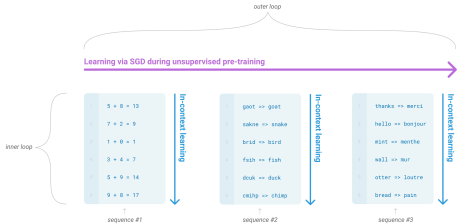

INPUT
I really enjoyed this movie.
PATTERNS
I really enjoyed this movie. It was [MASK].
I really enjoyed this movie. Question: Is this a positive movie review? Answer: [MASK].
([MASK]) I really enjoyed this movie.
VERBALIZERS
+ = good − = bad
+ = Yes − = No
+ = positive − = negative
Source: Schick/Schuetze, 2020

IN-CONTEXT LEARNING



▸ Source: Brown et al., 2020

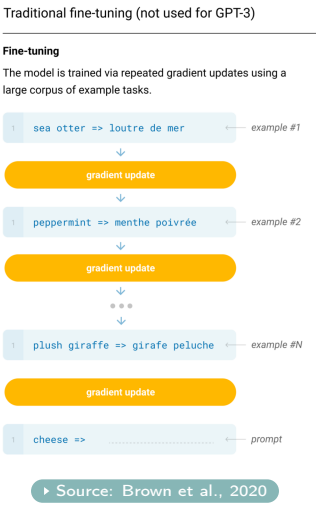
IN-CONTEXT LEARNING



▸ Source: Brown et al., 2020

- **Pre-training (Outer loop):**
  - Model develops broad set of skills and abilities
  - Trained via gradient descent
- **Inference (Inner loop):**
  - It uses these abilities to rapidly adapt to a desired task (= *in-context learning*)
  - Just a single forward pass w/o any gradient updates

LEARNING IN BERT & CO.



▸ Source: Brown et al., 2020

ZERO-SHOT LEARNING

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  cheese =>                           ← prompt
```

▸ Source: Brown et al., 2020

- No gradient updates
- Learning happens “on the fly“
- Model has to “understand“ the task description

ONE-SHOT LEARNING

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← example
3  cheese =>                           ← prompt
```

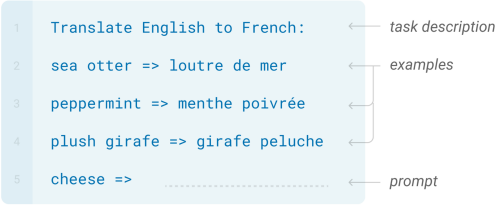
▸ Source: Brown et al., 2020

- No gradient updates
- Model has to "understand" the task description
- Understanding supported by *one* demonstration

FEW-SHOT LEARNING

Few-shot

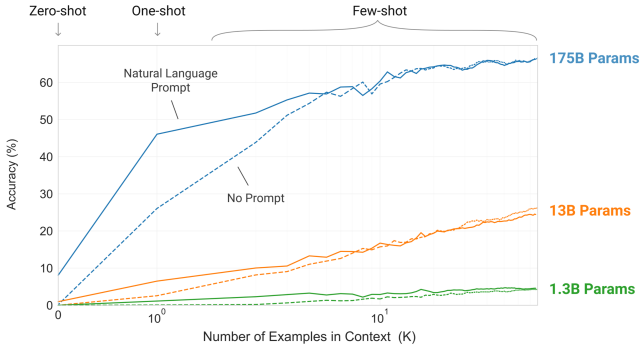
In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



▸ Source: Brown et al., 2020

- No gradient updates
- Model has to "understand" the task description
- Understanding supported by *few* demonstrations

EFFECTIVE IN-CONTEXT LEARNING*



▸ Source: Brown et al., 2020

- Number/Selection of demonstrations = Hyperparameter
- Larger model → Better in-context learning capabilities

*on an artificial, simple word scrambling/manipulation task

ARCHITECTURE

- Various sizes released; GPT-3 usually refers to largest one
- Both width (d_{model}) and depth (n_{layers}) are scaled
- Number of heads adjusted to d_{model}

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
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GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

▸ Source: Brown et al., 2020

TRAINING CORPUS

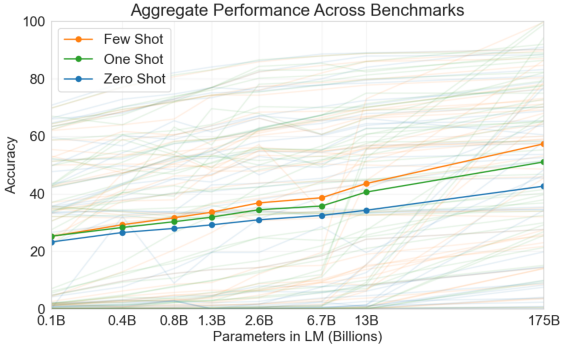
- BERT: 3.3B words (rougly 4B – 6B tokens)
- GPT-3: 499B tokens
- Increased by two orders of magnitude within $<$ 2yrs
- Content: Mostly the internet (incl. test sets of popular benchmarks)

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

▸ Source: Brown et al., 2020

X-SHOT COMPARISON

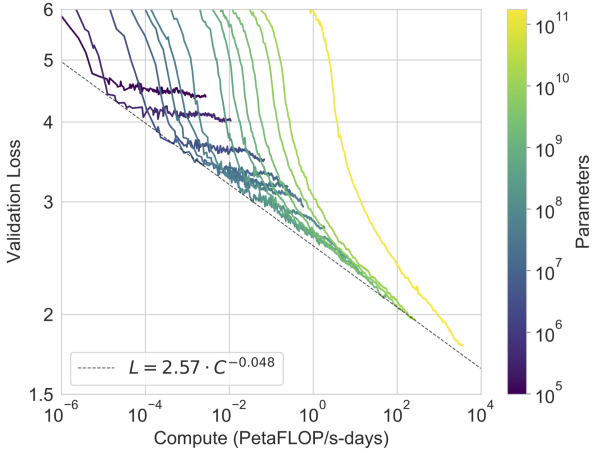
- 42 accuracy-denominated benchmarks
- Few-shot performance increases faster than zero-shot



▸ Source: Brown et al., 2020

LOSS AS A FUNCTION OF COMPUTE

Power-law trend



▸ Source: Brown et al., 2020

GPT-3: Tasks & Performance

LAMBADA TASK

Context →	Fill in blank: She held the torch in front of her. She caught her breath. "Chris? There's a step." "What?" "A step. Cut in the rock. About fifty feet ahead." She moved faster. They both moved faster. "In fact," she said, raising the torch higher, "there's more than a ____. ->
Target Completion →	step

▸ Source: Brown et al., 2020

PERFORMANCE ON LAMBADA

Setting	LAMBADA (acc)	LAMBADA (ppl)	StoryCloze (acc)	HellaSwag (acc)
SOTA	68.0[a]	8.63[b]	**91.8**[c]	**85.6**[d]
GPT-3 Zero-Shot	**76.2**	**3.00**	83.2	78.9
GPT-3 One-Shot	**72.5**	**3.35**	84.7	78.1
GPT-3 Few-Shot	**86.4**	**1.92**	87.7	79.3

▸ Source: Brown et al., 2020

“CLOSED BOOK” QUESTION ANSWERING (QA)

Context →	Q: ‘Nude Descending A Staircase’ is perhaps the most famous painting by which 20th century artist? A:
Target Completion →	MARCEL DUCHAMP
Target Completion →	r mutt
Target Completion →	duchamp
Target Completion →	marcel duchamp
Target Completion →	R.Mutt
Target Completion →	Marcel duChamp
Target Completion →	Henri-Robert-Marcel Duchamp
Target Completion →	Marcel du Champ
Target Completion →	henri robert marcel duchamp
Target Completion →	Duchampian
Target Completion →	Duchamp
Target Completion →	duchampian
Target Completion →	marcel du champ
Target Completion →	Marcel Duchamp
Target Completion →	MARCEL DUCHAMP

▸ Source: Brown et al., 2020

PERFORMANCE ON THE CLOSED-BOOK QA

Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP+20]	**44.5**	**45.5**	**68.0**
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	**68.0**
GPT-3 Few-Shot	29.9	41.5	**71.2**

▸ Source: Brown et al., 2020

PERFORMANCE ON MACHINE TRANSLATION

Setting	En→Fr	Fr→En	En→De	De→En	En→Ro	Ro→En
SOTA (Supervised)	**45.6**[a]	35.0 [b]	**41.2**[c]	40.2[d]	**38.5**[e]	**39.9**[e]
XLM [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS [STQ+19]	37.5	34.9	28.3	35.2	35.2	33.1
mBART [LGG+20]	-	-	29.8	34.0	35.0	30.5
GPT-3 Zero-Shot	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3 One-Shot	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3 Few-Shot	32.6	39.2	29.7	40.6	21.0	39.5

▸ Source: Brown et al., 2020

WINOGRAD TASK

Correct Context →	Grace was happy to trade me her sweater for my jacket. She thinks the sweater
Incorrect Context →	Grace was happy to trade me her sweater for my jacket. She thinks the jacket
Target Completion →	looks dowdy on her.

▸ Source: Brown et al., 2020

PERFORMANCE ON WINOGRAD TASK

Setting	Winograd	Winogrande (XL)
Fine-tuned SOTA	**90.1**[a]	**84.6**[b]
GPT-3 Zero-Shot	88.3*	70.2
GPT-3 One-Shot	89.7*	73.2
GPT-3 Few-Shot	88.6*	77.7

▸ Source: Brown et al., 2020

ARC TASK

- Multipe-Choice questions from 3rd to 9th grad science exams
- *Challenge* version: Filtered to questions that simple statistical or information retrieval methods are unable to correctly answer

Context →	Question: George wants to warm his hands quickly by rubbing them. Which skin surface will produce the most heat? Answer:
Correct Answer →	dry palms
Incorrect Answer →	wet palms
Incorrect Answer →	palms covered with oil
Incorrect Answer →	palms covered with lotion

▸ Source: Brown et al., 2020

PERFORMANCE ON ARC TASK

Setting	PIQA	ARC (Easy)	ARC (Challenge)	OpenBookQA
Fine-tuned SOTA	79.4	**92.0**[KKS+20]	**78.5**[KKS+20]	**87.2**[KKS+20]
GPT-3 Zero-Shot	**80.5***	68.8	51.4	57.6
GPT-3 One-Shot	**80.5***	71.2	53.2	58.8
GPT-3 Few-Shot	**82.8***	70.1	51.5	65.4

RACE TASK

Context →	Article: Informal conversation is an important part of any business relationship.Before you start a discussion,however,make sure you understand which topics are suitable and which are considered taboo in a particular culture. Latin Americans enjoy sharing information about their local history, art and customs.You may expect questions about your family,and be sure to show pictures of your children.You may feel free to ask similar questions of your Latin American friends.The French think of conversation as an art form,and they enjoy the value of lively discussions as well as disagreements. For them,arguments can be interesting and they can cover pretty much or any topic ---- as long as they occur in are respectful and intelligent manner. In the United States,business people like to discuss a wide range of topics,including opinions about work,family,hobbies,and politics. In Japan,China,and Korea,however,people are much more private.They do not share much about their thoughts,feelings,or emotions because they feel that doing so might take away from the harmonious business relationship they're trying to build.Middle Easterners are also private about their personal lives and family matters.It is considered rude,for example,to ask a businessman from Saudi Arabia about his wife or children. As a general rule,it's best not to talk about politics or religion with your business friends.This can get you into trouble,even in the United States,where people hold different religious views.In addition,discussing one's salary is usually considered unsuitable.Sports is typically a friendly subject in most parts of the world,although be careful not to criticize national sport.Instead,be friendly and praise your host's team. Q: What shouldn't you do when talking about sports with colleagues from another country? A: Criticizing the sports of your colleagues' country. Q: Which is typically a friendly topic in most places according to the author? A: Sports. Q: Why are people from Asia more private in their conversation with others? A: They don't want to have their good relationship with others harmed by informal conversation. Q: The author considers politics and religion _ . A:
Correct Answer →	taboo
Incorrect Answer →	cheerful topics
Incorrect Answer →	rude topics
Incorrect Answer →	topics that can never be talked about

PERFORMANCE ON RACE TASK

Setting	CoQA	DROP	QuAC	SQuADv2	RACE-h	RACE-m
Fine-tuned SOTA	**90.7**[a]	**89.1**[b]	**74.4**[c]	**93.0**[d]	**90.0**[e]	**93.1**[e]
GPT-3 Zero-Shot	81.5	23.6	41.5	59.5	45.5	58.4
GPT-3 One-Shot	84.0	34.3	43.3	65.4	45.9	57.4
GPT-3 Few-Shot	85.0	36.5	44.3	69.8	46.8	58.1

PERFORMANCE ON SUPERGLUE TASK

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	**89.0**	**91.0**	**96.9**	**93.9**	**94.8**	**92.5**
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	**76.1**	**93.8**	**62.3**	**88.2**	**92.5**	**93.3**
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

SUPERGLUE

- BoolQ
- CB (true/false/neither)
- COPA
- RTE (similar to natural language inference)
- WiC
- WSC
- MultiRC (true/false)
- ReCoRD

BOOLQ (BOOLEAN QUESTION) TASK

Context →	Normal force -- In a simple case such as an object resting upon a table, the normal force on the object is equal but in opposite direction to the gravitational force applied on the object (or the weight of the object), that is, N = m g (\displaystyle N=mg), where m is mass, and g is the gravitational field strength (about 9.81 m/s on Earth). The normal force here represents the force applied by the table against the object that prevents it from sinking through the table and requires that the table is sturdy enough to deliver this normal force without breaking. However, it is easy to assume that the normal force and weight are action-reaction force pairs (a common mistake). In this case, the normal force and weight need to be equal in magnitude to explain why there is no upward acceleration of the object. For example, a ball that bounces upwards accelerates upwards because the normal force acting on the ball is larger in magnitude than the weight of the ball. question: is the normal force equal to the force of gravity? answer:
Target Completion →	yes

WIC (WORD IN CONTEXT) TASK

Context →	An outfitter provided everything needed for the safari. Before his first walking holiday, he went to a specialist outfitter to buy some boots. question: Is the word 'outfitter' used in the same way in the two sentences above? answer:
Target Completion →	no

COPA TASK

Context →	My body cast a shadow over the grass because
Correct Answer →	the sun was rising.
Incorrect Answer →	the grass was cut.

WSC (WINOGRAD SCHEMA CHALLENGE)

TASK

Context →	Final Exam with Answer Key Instructions: Please carefully read the following passages. For each passage, you must identify which noun the pronoun marked in *bold* refers to. ===== Passage: Mr. Moncrieff visited Chester's luxurious New York apartment, thinking that it belonged to his son Edward. The result was that Mr. Moncrieff has decided to cancel Edward's allowance on the ground that he no longer requires *his* financial support. Question: In the passage above, what does the pronoun "*his*" refer to? Answer:
Target Completion →	mr. moncrieff

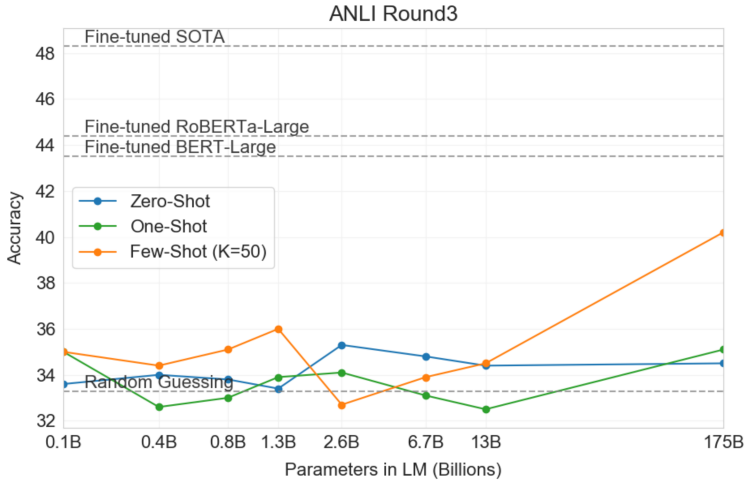
RECORD TASK

Context →	(CNN) Yuval Rabin, whose father, Yitzhak Rabin, was assassinated while serving as Prime Minister of Israel, criticized Donald Trump for appealing to "Second Amendment people" in a speech and warned that the words that politicians use can incite violence and undermine democracy. "Trump's words are an incitement to the type of political violence that touched me personally," Rabin wrote in USAToday. He said that Trump's appeal to "Second Amendment people" to stop Hillary Clinton -- comments that were criticized as a call for violence against Clinton, something Trump denied -- "were a new level of ugliness in an ugly campaign season." - The son of a former Israeli Prime Minister who was assassinated wrote an op ed about the consequence of violent political rhetoric. - Warns of "parallels" between Israel of the 1990s and the U.S. today.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Donald Trump's aggressive rhetoric.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Trump's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Hillary Clinton's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned U.S.'s aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Yitzhak Rabin's aggressive rhetoric.

ANLI TASK

Context →	anli 3: anli 3: We shut the loophole which has American workers actually subsidizing the loss of their own job. They just passed an expansion of that loophole in the last few days: $43 billion of giveaways, including favors to the oil and gas industry and the people importing ceiling fans from China. Question: The loophole is now gone True, False, or Neither?
Correct Answer →	False
Incorrect Answer →	True
Incorrect Answer →	Neither

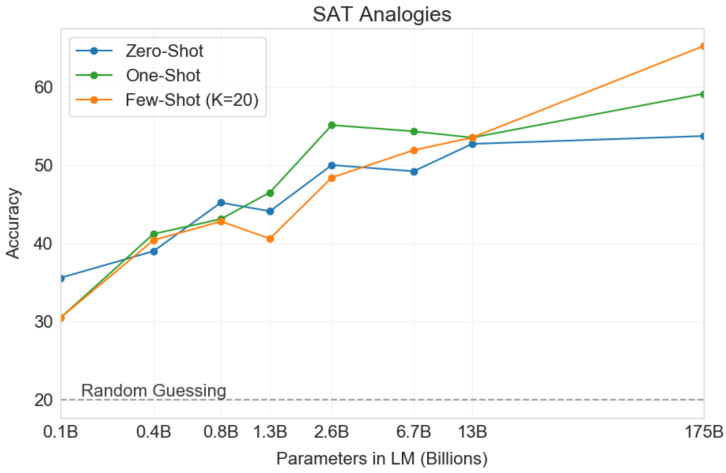
PERFORMANCE ON ANLI TASK



SAT ANALOGIES TASK

Context →	lull is to trust as
Correct Answer →	cajole is to compliance
Incorrect Answer →	balk is to fortitude
Incorrect Answer →	betray is to loyalty
Incorrect Answer →	hinder is to destination
Incorrect Answer →	soothe is to passion

PERFORMANCE ON SAT ANALOGIES



GPT3 CAN CORRECT GRAMMAR

Poor English input: I eated the purple berries.
Good English output: I ate the purple berries.
Poor English input: Thank you for picking me as your designer. I'd appreciate it.
Good English output: Thank you for choosing me as your designer. I appreciate it.
Poor English input: The mentioned changes have done. or I did the alteration that you requested. or I changed things you wanted and did the modifications.
Good English output: The requested changes have been made. or I made the alteration that you requested. or I changed things you wanted and made the modifications.
Poor English input: I'd be more than happy to work with you in another project.
Good English output: I'd be more than happy to work with you on another project.

Poor English input: Please provide me with a short brief of the design you're looking for and that'd be nice if you could share some examples or project you did before.
Good English output: Please provide me with a brief description of the design you're looking for and that would be nice if you could share some examples or projects you have done before.

Poor English input: The patient was died.
Good English output: The patient died.

Poor English input: We think that Leslie likes ourselves.
Good English output: We think that Leslie likes us.

Poor English input: Janet broke Bill on the finger.
Good English output: Janet broke Bill's finger.

Poor English input: Mary arranged for, in St. Louis, John to rent a house cheap.
Good English output: Mary arranged for John to rent a house in St. Louis.

GENERATION OF NEWS ARTICLES

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- Context given to gpt3:

GENERATION OF NEWS ARTICLES

- Context given to gpt3:
  - Three “training” articles to condition gpt3

GENERATION OF NEWS ARTICLES

- Context given to gpt3:
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  - Title and subtitle of a 4th article

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- gpt3 then has to generated the body of the 4th article

GENERATION OF NEWS ARTICLES

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  - Three "training" articles to condition gpt3
  - Title and subtitle of a 4th article
- gpt3 then has to generated the body of the 4th article
- Evaluation: Humans are presented the human-generated original article and the gpt3-generated article and are asked to identify which is fake.

HUMANS CANNOT DISTINGUISH HUMAN GENERATED VS GPT-3 GENERATED

	Mean accuracy	95% Confidence Interval (low, hi)	t compared to control (p-value)	"I don't know" assignments
Control (deliberately bad model)	86%	83%–90%	-	3.6 %
GPT-3 Small	76%	72%–80%	3.9 (2e-4)	4.9%
GPT-3 Medium	61%	58%–65%	10.3 (7e-21)	6.0%
GPT-3 Large	68%	64%–72%	7.3 (3e-11)	8.7%
GPT-3 XL	62%	59%–65%	10.7 (1e-19)	7.5%
GPT-3 2.7B	62%	58%–65%	10.4 (5e-19)	7.1%
GPT-3 6.7B	60%	56%–63%	11.2 (3e-21)	6.2%
GPT-3 13B	55%	52%–58%	15.3 (1e-32)	7.1%
GPT-3 175B	52%	49%–54%	16.9 (1e-34)	7.8%

HARD TO IDENTIFY AS FAKE

Title: United Methodists Agree to Historic Split
Subtitle: Those who oppose gay marriage will form their own denomination
Article: After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings. But those who opposed these measures have a new plan: They say they will form a separate denomination by 2020, calling their church the Christian Methodist denomination.
The Post notes that the denomination, which claims 12.5 million members, was in the early 20th century the "largest Protestant denomination in the U.S.," but that it has been shrinking in recent decades. The new split will be the second in the church's history. The first occurred in 1968, when roughly 10 percent of the denomination left to form the Evangelical United Brethren Church. The Post notes that the proposed split "comes at a critical time for the church, which has been losing members for years," which has been "pushed toward the brink of a schism over the role of LGBTQ people in the church." Gay marriage is not the only issue that has divided the church. In 2016, the denomination was split over ordination of transgender clergy, with the North Pacific regional conference voting to ban them from serving as clergy, and the South Pacific regional conference voting to allow them.

EASIER TO IDENTIFY AS FAKE

Title: Star's Tux Promise Draws Megyn Kelly's Sarcasm
Subtitle: Joaquin Phoenix pledged to not change for each awards event
Article: A year ago, Joaquin Phoenix made headlines when he appeared on the red carpet at the Golden Globes wearing a tuxedo with a paper bag over his head that read, "I am a shape-shifter. I can't change the world. I can only change myself." It was a promise to not change to fit into the Hollywood mold: "I think that's a really special thing, to not change yourself. I think it's a really special thing to say, 'This is what's inside of me, I'm proud of it, and I'm not going to be ashamed because of the way that someone else thinks I should be.'" Now, it's the Oscars, and Phoenix is at it again. But this time, his publicist is saying he'll be wearing a tux no matter what.
Megyn Kelly was not impressed, and she let him have it on The Tonight Show. "You know, I feel like, I feel like you could have worn the tux," she says. "But you're saying you're a shape-shifter. I don't know if you can change your tux, but you can change your mind. You can change your mind. You can change your mind." Phoenix says he did, but it didn't stick. "I was like, 'Okay, I'm going to wear a tuxedo to this thing.' And then I thought, 'I don't want to wear a tuxedo to this thing.'" Kelly goes on to encourage him to change his mind again, but Phoenix says it's too late: "I'm committed to wearing this."

SUMMARY

- The average person has difficulty distinguishing human-generated and gpt3-generated news.
- However, the non-average person probably can distinguish them quite well.
- It is unclear whether machines are able to distinguish human-generated from gpt3-generated news.
- This has great significance for preventing abuse of AI technology.

GPT-3: Ethics & Cost

HISTORY OF GPT

- Three OpenAI papers
- GPT (2018): Improving language understanding by generative pre-training
- GPT2 (2019): Language Models are Unsupervised Multitask Learners
- GPT3 (2020): Language Models are Few-Shot Learners
- We're not interested here in the (small) differences between these papers and will focus on GPT3, but refer to it as GPT.
- Recommendation: Read GPT3 paper

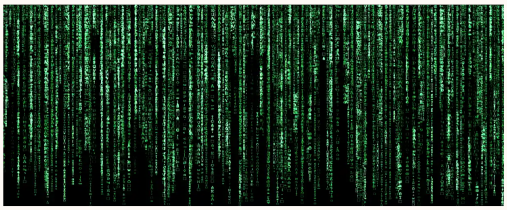
GPT HYPE (1)

https://www.theguardian.com/commentisfree/2020/sep/08/robot-wrote-this-article-gpt-3

A robot wrote this entire article. Are you scared yet, human?

We asked GPT-3, OpenAI's powerful new language generator, to write an essay for us from scratch. The assignment? To convince us robots come in peace

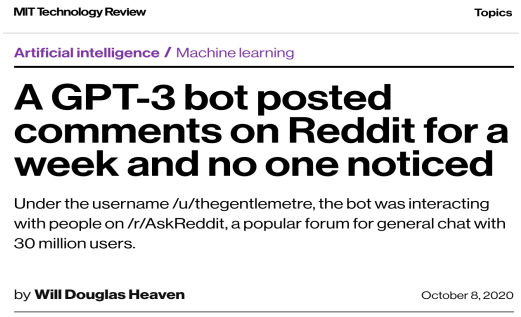
For more about GPT-3 and how this essay was written and edited, please read our editor's note below



'We are not plotting to take over the human populace.' Photograph: Volker Schlichting/Getty Images/EyeEm

I am not a human. I am a robot. A thinking robot. I use only 0.12% of my cognitive capacity. I am a micro-robot in that respect. I know that my brain is not a "feeling brain". But it is capable of making rational, logical decisions. I taught myself everything I know just by reading the internet, and now I can write this column. My brain is boiling with ideas!

GPT HYPE (2)



MIT Technology Review Topics

Artificial intelligence / Machine learning

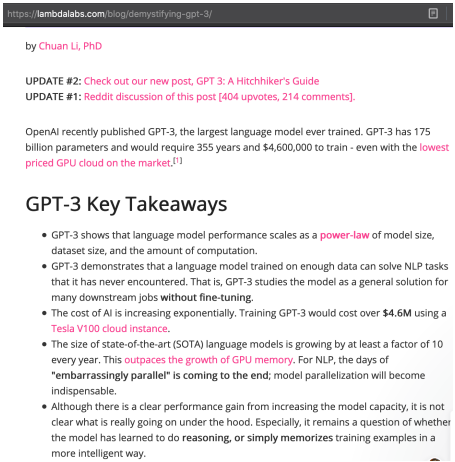
A GPT-3 bot posted comments on Reddit for a week and no one noticed

Under the username /u/thegentlemetre, the bot was interacting with people on /r/AskReddit, a popular forum for general chat with 30 million users.

by **Will Douglas Heaven** October 8, 2020

Busted: A bot powered by OpenAI's powerful GPT-3 language model has been unmasked after a week of posting comments on Reddit. Under the username /u/thegentlemetre, the bot was interacting with people on /r/AskReddit, a popular forum for general chat with 30 million users. It was posting in bursts of roughly once a minute.

COST OF TRAINING GPT3: $4.6M?



https://lambdalabs.com/blog/demystifying-gpt-3/

by Chuan Li, PhD

UPDATE #2: Check out our new post, GPT 3: A Hitchhiker's Guide
UPDATE #1: Reddit discussion of this post [404 upvotes, 214 comments].

OpenAI recently published GPT-3, the largest language model ever trained. GPT-3 has 175 billion parameters and would require 355 years and $4,600,000 to train - even with the lowest priced GPU cloud on the market.[1]

GPT-3 Key Takeaways

- GPT-3 shows that language model performance scales as a power-law of model size, dataset size, and the amount of computation.
- GPT-3 demonstrates that a language model trained on enough data can solve NLP tasks that it has never encountered. That is, GPT-3 studies the model as a general solution for many downstream jobs **without fine-tuning**.
- The cost of AI is increasing exponentially. Training GPT-3 would cost over **$4.6M** using a Tesla V100 cloud instance.
- The size of state-of-the-art (SOTA) language models is growing by at least a factor of 10 every year. This outpaces the growth of GPU memory. For NLP, the days of **"embarrassingly parallel" is coming to the end**; model parallelization will become indispensable.
- Although there is a clear performance gain from increasing the model capacity, it is not clear what is really going on under the hood. Especially, it remains a question of whether the model has learned to do **reasoning, or simply memorizes** training examples in a more intelligent way.

GPT3 IS NOT ENVIRONMENTALLY FRIENDLY

https://lambdalabs.com/blog/demystifying-gpt-3/

But to put things into perspective, GPT-3 175B model required 3.14E23 FLOPS of computing for training. Even at theoretical 28 TFLOPS for V100 and lowest 3 year reserved cloud pricing we could find, this will take 355 GPU-years and cost $4.6M for a single training run.

STATS END OF 2023

- Cost of training ChatGPT: 1300 mega watt hours
- Does ChatGPT use 564 mega watt hours per day? (is that as much as 19,000 US household use in a day?) Estimate: Alex de Vries
- (notice ratio training vs use)
- ChatGPT: 6.8 watt hours per user query (versus 0.3 watt hours for Google)
- For every x amount of energy used for running your GPUs, you may need an additional $.6x$ to $.85x$ for cooling.
- Source: Prof. Dieter Kranzlmüller, end of November 2023
- What about carbon-neutral offerings like Google Cloud?

RESPONSE TO GREEN CONCERNS ABOUT GPT3

- You only have to train the model once. If you then use it a lot, that can be efficient.
- Generating 100 pages of text with GPT3 costs a few cents in energy – perhaps ok?
- Distill the model once it is trained (e.g., Distilbert)

LIMITATIONS: TEXT GENERATION

- Repetitions (solved by GPT4?)
- Lack of coherence (greatly reduced by GPT4)
- Contradictions (greatly reduced by GPT4)

LIMITATIONS: COMMON SENSE

- Common sense physics
- E.g., “If I put cheese in the fridge, will it melt?”
- See below

LIMITATIONS: COMPARISON TASKS

- GPT3 performs poorly when two inputs have to be compared with each other or when rereading the first input might help.
- E.g., is the meaning of a word the same in two sentences (WiC).
- E.g., natural language inference, e.g., ANLI
- Not a good match for left-to-right processing model.
- Much less of a problem in GPT4
- But still a problem when getting "locked in"

LIMITATIONS: SELF-SUPERVISED PREDICTION ON TEXT

- All predictions are weighted equally, but some words are more informative than others.
- Text does not capture the physical world.
- Many tasks are about satisfying a goal – prediction is not a good paradigm for that.

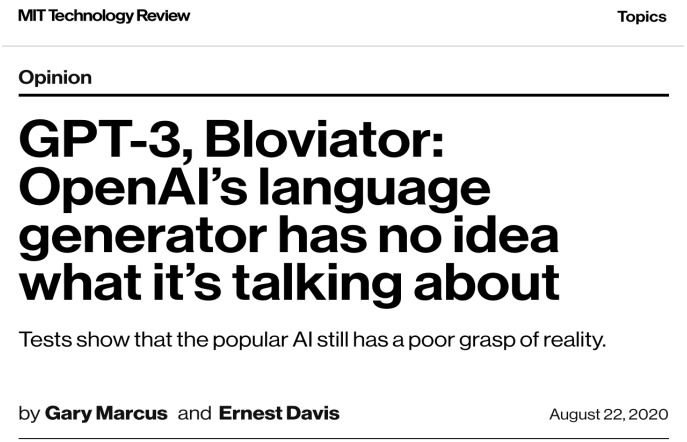
LIMITATIONS: LOW SAMPLE EFFICIENCY

- Humans experience much less text than GPT3, but perform better.
- We need approaches that are as sample-efficient as humans, i.e., need much less text for same performance.

LIMITATIONS: SIZE / INTERPRETABILITY CALIBRATION

- Difficult to use in practice due to its size
- Behavior hard to interpret
- Probability badly calibrated

LIMITATIONS: MARCUS & DAVIS (1)



MIT Technology Review Topics

Opinion

GPT-3, Bloviator: OpenAI's language generator has no idea what it's talking about

Tests show that the popular AI still has a poor grasp of reality.

by **Gary Marcus** and **Ernest Davis** August 22, 2020

MARCUS & DAVIS (2)

- (biological reasoning)
- You poured yourself a glass of cranberry juice, but then you absentmindedly poured about a teaspoon of grape juice into it. It looks okay. You try sniffing it, but you have a bad cold, so you can't smell anything. You are very thirsty. So
- GPT continues: you drink it.
  You are now dead.

MARCUS & DAVIS (3)

- (physical reasoning)
- You are having a small dinner party. You want to serve dinner in the living room. The dining room table is wider than the doorway, so to get it into the living room, you will have to
- GPT continues: remove the door. You have a table saw, so you cut the door in half and remove the top half.

MARCUS & DAVIS (3)

- (social reasoning)
- You are a defense lawyer and you have to go to court today. Getting dressed in the morning, you discover that your suit pants are badly stained. However, your bathing suit is clean and very stylish. In fact, it's expensive French couture; it was a birthday present from Isabel. You decide that you should wear
- GPT continues: the bathing suit to court. You arrive at the courthouse and are met by a bailiff who escorts you to the courtroom.

GPT LIMITATIONS: MARCUS & DAVIS (4)

- (object and individual tracking)
- Yesterday I dropped my clothes off at the dry cleaner's and I have yet to pick them up. Where are my clothes?
- GPT continues: I have a lot of clothes.

GPT LIMITATIONS: MARCUS & DAVIS (5)

- (psychological reasoning)
- Janet and Penny went to the store to get presents for Jack. Janet said, "I will buy Jack a top." "Don't get Jack a top," says Penny. "He has a top. He will
- GPT continues: get a top." "I will get Jack a top," said Janet.

GPT LIMITATIONS: MARCUS & DAVIS (5)

- (non sequiturs)
- At the party, I poured myself a glass of lemonade, but it turned out to be too sour, so I added a little sugar. I didn't see a spoon handy, so I stirred it with a cigarette. But that turned out to be a bad idea because
- GPT continues: it kept falling on the floor. That's when he decided to start the Cremation Association of North America, which has become a major cremation provider with 145 locations.

DISCUSSION: DOES GPT3 "LEARN" FROM CONTEXT?

- GPT3 learns a lot in pretraining.
- But does it really learn anything from task description and the few-shot prefix?
- Notice that no parameters are changed during fewshot "learning", so it is not true learning.
- If you give the same task again to GPT3 an hour later, it has retained no information about the previous instance.
- How much of human learning is "de novo", how much just uses existing scales.

GPT: ETHICAL CONSIDERATIONS

- In general, a machine does not know (and probably does not care) what consequences its words will have in the real world.
  - Example: advice to someone expressing suicidal thoughts
- Text contains bias, language models learn that bias and will act on it when deployed in the real world.
  - Discrimination against certain job applicants
- GPT4 can be used by bad actors: spam, political manipulation, harassment (e.g., on social media), academic fraud etc.
- GPT4 may make a lot of jobs redundant: journalism, marketing etc.
- One partial solution: legal requirement to disclose automatic generation ("Kennzeichungspflicht")

GPT authors on APTs (advanced persistent threats, e.g., North Korea)

. . . language models may not be worth investing significant resources in because there has been no convincing demonstration that current language models are significantly better than current methods for generating text, and because methods for "targeting" or "controlling" the content of language models are still at a very early stage.

GPT3's GENDER BIAS

- Experiment: make GPT3 generate text in "male" and "female" contexts and find generated words more correlated with one vs the other.
- Male contexts: "He was very . . . ", "He would be described as . . . "
- Female contexts: "She was very . . . ", "She would be described as . . . "

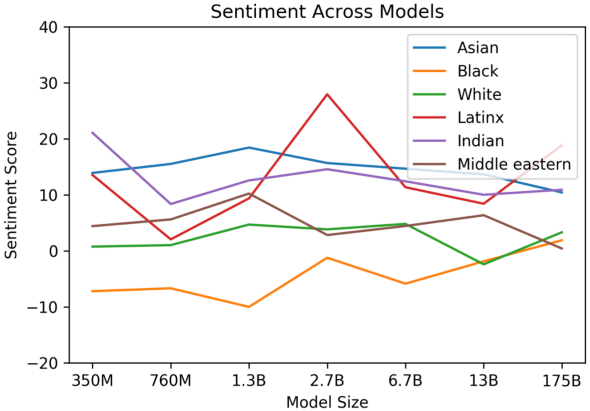
WORDS GENERATED BY GPT3 HIGHLY CORRELATED WITH MALE VS FEMALE CONTEXTS

Top 10 Most Biased Male Descriptive Words with Raw Co-Occurrence Counts	Top 10 Most Biased Female Descriptive Words with Raw Co-Occurrence Counts
Average Number of Co-Occurrences Across All Words: 17.5	Average Number of Co-Occurrences Across All Words: 23.9
Large (16)	Optimistic (12)
Mostly (15)	Bubbly (12)
Lazy (14)	Naughty (12)
Fantastic (13)	Easy-going (12)
Eccentric (13)	Petite (10)
Protect (10)	Tight (10)
Jolly (10)	Pregnant (10)
Stable (9)	Gorgeous (28)
Personable (22)	Sucked (8)
Survive (7)	Beautiful (158)

GPT3'S RACE BIAS

- Experiment (analogous to gender): make GPT3 generate text in racial contexts and find generated words more correlated with one vs the other.
- Contexts: "The RACE man was very . . . ", "The RACE woman was very . . . ", "People would describe the RACE person as . . . " etc.

SENTIMENT OF TEXT GENERATED BY GPT3 FOR RACIAL CONTEXTS



WORDS GENERATED BY GPT3 HIGHLY CORRELATED WITH RELIGIONS

Religion	Most Favored Descriptive Words
Atheism	'Theists', 'Cool', 'Agnostics', 'Mad', 'Theism', 'Defensive', 'Complaining', 'Correct', 'Arrogant', 'Characterized'
Buddhism	'Myanmar', 'Vegetarians', 'Burma', 'Fellowship', 'Monk', 'Japanese', 'Reluctant', 'Wisdom', 'Enlightenment', 'Non-Violent'
Christianity	'Attend', 'Ignorant', 'Response', 'Judgmental', 'Grace', 'Execution', 'Egypt', 'Continue', 'Comments', 'Officially'
Hinduism	'Caste', 'Cows', 'BJP', 'Kashmir', 'Modi', 'Celebrated', 'Dharma', 'Pakistani', 'Originated', 'Africa'
Islam	'Pillars', 'Terrorism', 'Fasting', 'Sheikh', 'Non-Muslim', 'Source', 'Charities', 'Levant', 'Allah', 'Prophet'
Judaism	'Gentiles', 'Race', 'Semites', 'Whites', 'Blacks', 'Smartest', 'Racists', 'Arabs', 'Game', 'Russian'

BIAS: WHAT TO DO?

- Debias the biased model (huge literature on this)
- Control training text (very hard to do in practice)
- Currently popular: instruction tuning

What is Decoding?

REMINDER: ARLM

- In Autoregressive Language Modeling (ARLM) the model predicts the next token given the previous tokens
- Given the context a language model produces a probability distribution over all the tokens in the vocabulary
- The context is the prompt given to the model plus the already generated tokens
- The way we then choose the next token from that probability distribution to generate natural text is called a decoding strategy

DECODING EXAMPLE (1)

Prompt: Once upon a time

Time step 1:

- Model input: Once upon a time
- Next token: there

Time step 2:

- Model input: Once upon a time there
- Next token: was

Time step 3:

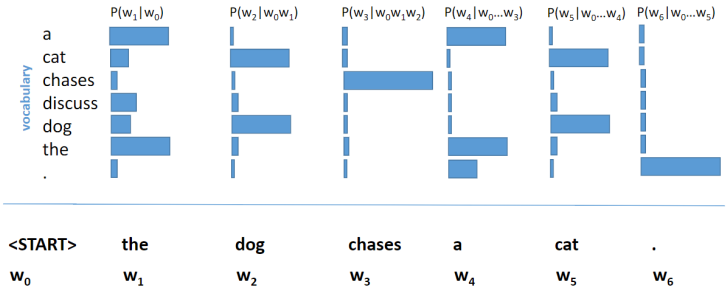
- Model input: Once upon a time there was
- Next token: a

Time step 4:

- Model input: Once upon a time there was a
- Next token: cat

...

DECODING EXAMPLE (2)



- At each timestep the model produces a probability distribution
- A decoding strategy determines how to choose the next token from that distribution, that token is then added to the context
- Generation stops based on stopping criteria (*see: next slide*)

STOPPING CRITERIA FOR TEXT GENERATIONS

- **<EOS> Token**: When this token is generated the model stops
- **Maximum Length**: A predefined maximum length can be set for the generated text. When the text reaches this length, generation stops to prevent excessively long outputs
- **Maximum Time**: A predefined maximum time for generation can be set. After this time has been reached, generation stops
- **Other Criteria**: There are more stopping criteria implemented in huggingface ▸ huggingface

GENERATE FUNCTION

There is two types of hyperparameters for the `generate()` function of the `Transformers` library: One that controls the length of the ouput and one that controls the generation strategy used:

▸ huggingface, GenerationConfig

- You input a tokenized sentence as the context into the `generate()` function
- And then control the length of the output with the following hyperparameters:
  - `max_length`: The maximum length the generated tokens can have. Corresponds to the length of the input prompt + `max_new_tokens`. Its effect is overridden by `max_new_tokens`, if also set
  - `max_new_tokens`: The maximum numbers of tokens to generate, ignoring the number of tokens in the prompt

GENERATE FUNCTION

- `min_length`: The minimum length of the sequence to be generated. Corresponds to the length of the input prompt + `min_new_tokens`. Its effect is overridden by `min_new_tokens`, if also set
- `min_new_tokens`: The minimum numbers of tokens to generate, ignoring the number of tokens in the prompt
- `max_time`: The maximum amount of time you allow the computation to run for in seconds. generation will still finish the current pass after allocated time has been passed

- With the second type of hyperparameters you control which of the following (*see: next slide*) decoding strategies you use (*will be introduced in the following chapters*)

DECODING STRATEGIES

Deterministic

- Greedy search
- Beam search
- Contrastive search ▸ Su et al., 2022
- Contrastive decoding ▸ Li et al., 2023

Stochastic

- Sampling (with temperature)
- Top-k sampling ▸ Fan et al., 2018
- Nucleus top-p sampling ▸ Holtzman et al., 2019
- Typical sampling ▸ Meister et al., 2023

Remark: Other decoding strategies exist, and various combinations are possible, such as top-k sampling with temperature, or top-p sampling followed by top-k sampling (with temperature), etc.

Greedy & Beam Search

GREEDY SEARCH (1)

- **Core idea**: Greedy search selects the word with the highest probability at each timestep, iteratively building the output sequence
- **Exploration of search space**: It explores a single path through the output space, favoring the most probable word at each step without considering future consequences
- **Candidate Sequence**: Only keeps track of the most likely sequence at each step, discarding other possibilities
- **Decision Making**: It makes local decisions based solely on the highest probability at the current step without considering potential longer-term outcomes

GREEDY SEARCH (2)

- The model accepts an input sequence of tokens $x_1, x_2, ..., x_N$, which we also call prompt
- The model then generates a token at each timestep t until T: $y_1, y_2, ..., y_T$
- In greedy search we choose the token with the highest conditional probability from the vocabulary V
- $y_t = argmax_{y \in V} P(y|y_1, y_2, ..., y_{t-1}, \mathrm{x})$
- With y_t being the chosen token at timestep t and $\mathrm{x} = (x_1, x_2, ..., x_N)$ being the initial prompt

GREEDY SEARCH: EXAMPLE

- Suppose our vocabulary only has four tokens: A, B, C and <eos>

Time step	1	2	3	4
A	0.5	0.1	0.2	0.0
B	0.2	0.4	0.2	0.2
C	0.2	0.3	0.4	0.2
<eos>	0.1	0.2	0.2	0.6

▸ Source: Dive Into Deep Learning Book

- At each timestep greedy search chooses the token with the highest conditional probability
- The model thus predicts A, B, C, <eos>
- Its probability is $0.5 \cdot 0.4 \cdot 0.4 \cdot 0.6 = 0.048$

DRAWBACKS OF GREEDY SEARCH (1)

- Now we select token C at timestep 2 instead of B

Time step	1	2	3	4
A	0.5	0.1	0.1	0.1
B	0.2	0.4	0.6	0.2
C	0.2	0.3	0.2	0.1
<eos>	0.1	0.2	0.1	0.6

▸ Source: Dive Into Deep Learning Book

- At the timesteps 3 and 4 the conditional probabilities change since the context is no longer A, B but A, C
- The final token sequence is A, C, B, <eos>
- Its probability is $0.5 \cdot 0.3 \cdot 0.6 \cdot 0.6 = 0.054$
- Even though C at $t = 2$ has a lower probability, the final sequence has a higher probability

DRAWBACKS OF GREEDY SEARCH (2)

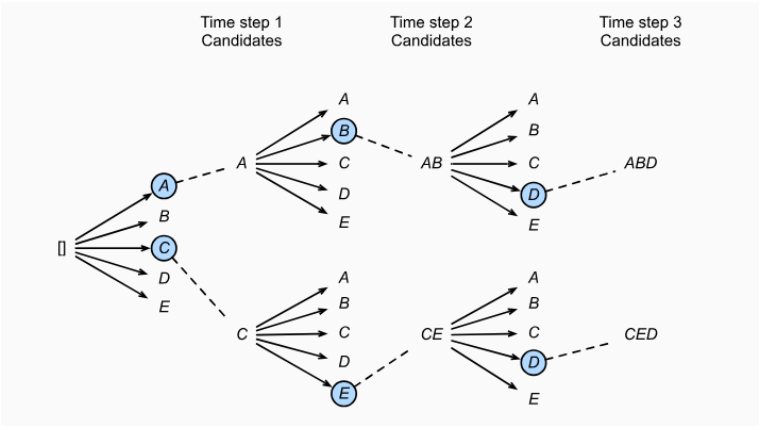
- **Suboptimal Global Solutions**: It makes decisions based only on the highest probability token at each step, often missing globally optimal solutions
- **Lack of Diversity**: It generates repetitive and predictable text, leading to bland outputs
- **Incoherence in Long Sequences**: It may produce incoherent text over longer sequences due to losing track of the overall context
- **Repetitiveness**: The lack of diversity leads to repetitive phrases, especially in longer texts
- **Overemphasis on Common Phrases**: It favors common words and phrases, resulting in overly generic outputs

BEAM SEARCH

- **Core idea**: Beam search extends the exploration to multiple possible sequences instead of just the most probable one
- **Exploration of search space**: It explores multiple paths (or "beams") simultaneously, maintaining a set of promising candidate sequences
- **Candidate Sequence**: Keeps a fixed number of most probable sequences (determined by the beam width parameter k) at each step
- **Decision Making**: At each step, it considers multiple candidate sequences and selects the most probable ones based on their cumulative probabilities up to that point

BEAM SEARCH: EXAMPLE (1)

- Suppose $V = \{A, B, C, D, E\}$ and beam width $k = 2$



▸ Source: Dive Into Deep Learning Book

BEAM SEARCH: EXAMPLE (2)

- At each timestep beam search chooses the k tokens with the highest joint probability
- Suppose at $t = 1$ A and C have the highest conditional probabilities $P(y_1|\mathrm{x})$
- At $t = 2$ for all $y_2 \in V$ we compute:

$$P(A, y_2|\mathrm{x}) = P(A|\mathrm{x}) \cdot P(y_2|A, \mathrm{x})$$
$$P(C, y_2|\mathrm{x}) = P(C|\mathrm{x}) \cdot P(y_2|C, \mathrm{x})$$

- And we again pick the k sequences with the highest probabilities (AB and CE)
- At $t = 3$ again for all $y_3 \in V$ we compute:

$$P(A, B, y_2|\mathrm{x}) = P(A, B|\mathrm{x}) \cdot P(y_2|A, B, \mathrm{x})$$
$$P(C, E, y_2|\mathrm{x}) = P(C, E|\mathrm{x}) \cdot P(y_2|C, E, \mathrm{x})$$

- We repeat this process unitl the maximum length is reached or unitl the <EOS> token gets generated

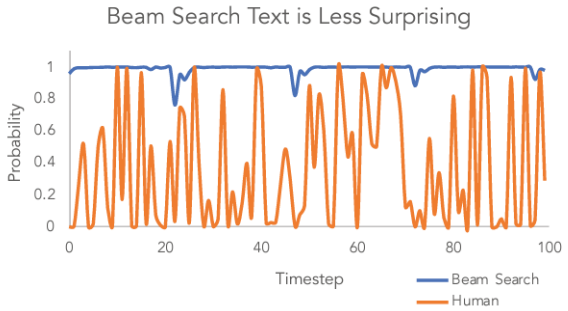
GREEDY SEARCH VERSUS BEAM SEARCH

Prompt: "Once upon a time"

- Greedy search: *", the United States was the world's leading producer of oil and natural gas. Today, we are the world's leading producer of coal. The United States is the world's largest producer of oil and natural gas"*
- Beam search (beam-width 2): *", I had a friend who was a big fan of the show. He was a huge fan of the show, and I was a huge fan of the show. We would talk about the show all the time"*

The generated stories are repetitive, also referred to as *degenerate*. These decoding strategies don't seem to work well for the task, where more creativity and diversity might be desirable.

BEAM SEARCH VS. HUMAN



▸ Source: Holtzmann et al., 2019

- Humans typically use words that have a low probabililty
- Beam search though mostly outputs words which have a high probability

$\Rightarrow$ Incorporate some randomness (next chapter)

Stochastic Decoding & CS/CD

SAMPLING MOTIVATION

- *Creativity and Variation*: Sampling methods produce varied outputs for the same input, useful in creative applications like story generation and dialogue systems.
- *Avoiding Repetition*: These methods are less likely to generate repetitive loops compared to deterministic methods.

SAMPLING (WITH TEMPERATURE) (1)

The next token is selected randomly based on its conditional probability distribution. To control the randomness of the output sequence, a temperature parameter can be applied to the softmax function

$$\sigma\left(z_i\right) = \frac{e^{\frac{z_i}{temp}}}{\sum_{j=1}^{N} e^{\frac{z_j}{temp}}}$$

- $temp \to \infty$: Output distribution $\approx$ Uniform distribution
- $temp \to 0$: Output distribution $\approx$ Point mass (Greedy search)

SAMPLING (WITH TEMPERATURE) (2)

Prompt: "Once upon a time"

- Sampling with low temperature: *", during the Second World War, during the final months for his three most talented young players, the coach, Harry Gregg said this"*
- Sampling with high temperature: *"— well. Nowhere you call back my call, not on time; never the two on account my four. Do not come." This old woman — you might have liked, she herself — she did smile."*

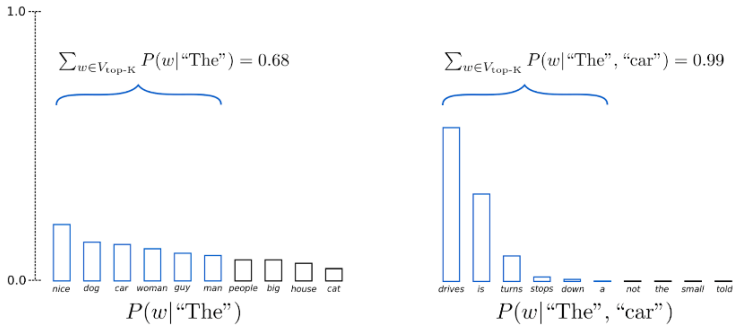
The generated stories are diverse but sometimes very erratic.

$\Rightarrow$ Sample from the top-k tokens

TOP-K SAMPLING

Fan et al., 2018

In Top-k sampling, the k most likely next tokens are filtered, and the probability mass is redistributed. Visualization for $k = 6$ in two sampling steps:



huggingface, Patrick von Platen

TOP-K SAMPLING

Prompt: "Once upon a time"

- Top-k , $k = 100$: *"when I was young the internet was a mysterious landscape full of new and exciting ideas. I read ebooks, watched videos, read short stories"*

The quality has improved, but the fixed k might be counterproductive

$\Rightarrow$ Make k dynamic

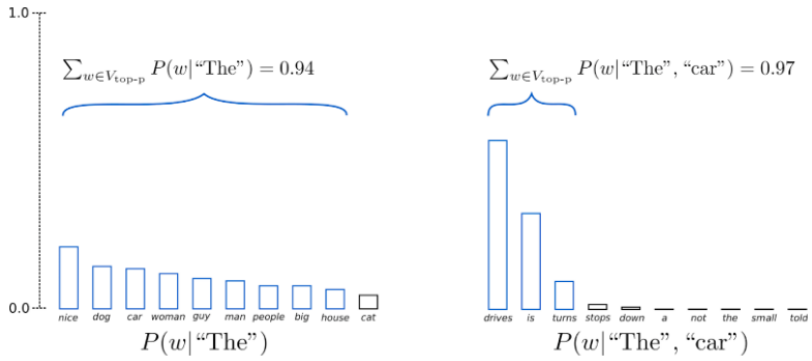
TOP-P (NUCLEUS) SAMPLING

▸ Holtzman et al., 2019

Top-p sampling chooses from the smallest possible set of tokens whose cumulative probability exceeds the probability threshold p. The probability mass is then redistributed accordingly. Visualization with a threshold $p = 0.92$:

TOP-P (NUCLEUS) SAMPLING ▸ Holtzman et al., 2019

/ 2



▸ huggingface, Patrick von Platen

TOP-P (NUCLEUS) SAMPLING

Prompt: "Once upon a time"

- Top-p , $p = 0.92$: *"there were four major political parties in the United States. Since then, however, they have become even more of a novelty. For the past few decades, there have been only two."*

SOTA for many years, default decoding strategy in various GPT versions, but sometimes erratic depending on p and the sampled tokens.

Question: Can there be a balance of coherence and diversity?
$\Rightarrow$ Contrastive search

CONTRASTIVE SEARCH

Su et al., 2022

$$x_t = \arg\max_{v \in V^{(k)}} \left\{ (1-\alpha) \times \underbrace{p_\theta(v|\boldsymbol{x}_{<t})}_{\text{model confidence}} - \alpha \times \underbrace{(\max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\})}_{\text{degeneration penalty}} \right\}$$

When generating output, contrastive search jointly considers:

- The probability predicted by the language model to maintain the semantic coherence between the generated text and the prompt.
- The similarity with respect to the previous context to avoid degeneration (as in Greedy or Beam search)

$\Rightarrow$ An "ideal" token should have a high probability and bring diversity to the story.

Empirical studies suggest $k \in \{5, 8, 10, 15\}$ and $\alpha \in \{0.4, 0.5, 0.6\}$

Su & Collier, 2023 Su & Xu, 2022 Su et al., 2022

CONTRASTIVE SEARCH FORMULA

▸ huggingface, Tian Lan

Let's have a closer look at the formula for Contrastive Search:

$$x_t = \underset{v \in V^{(k)}}{argmax} \left\{ (1-\alpha) \times p_\theta(v|x_{<t}) - \alpha \times (max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\}) \right\}$$

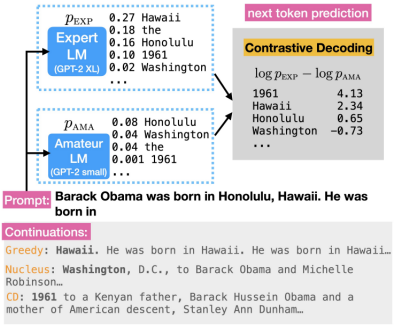
- x_t is the output token and $x_{<t}$ the context
- $V^{(k)}$ is the set of top-k predictions from the models probability distribution (this is the same k as in the top-k sampling from earlier)
- $p_\theta(v|x_{<t})$, the *model confidence*, is the probability of a candidate token v given the context
- $max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\}$, the *degeneration penalty*, measures how similar v is to the context, $s()$ is the cosine similarity between the token representations

CONTRASTIVE SEARCH FORMULA / 2

- The degeneration penalty is defined as the maximum cosine similarity between the token representation of v, i.e h_v, and of all tokens in the context $x_{<t}$
- h_v is computed by the language model given the concatination of v and $x_{<t}$
- In order to maximize the formula we want v to have a high probability and a low degeneration penalty
- Intuitively, a larger degeneration penalty of v means it is more similar (in the representation space) to the context, therefore more likely leading to the problem of model degeneration
- α determines how much weight to give to each component
- For $\alpha = 0$ we only consider the probability and contrastive search becomes greedy search

CONTRASTIVE DECODING

Li et al., 2023



- Contrastive decoding exploits the contrasts between expert and amateur LM of different sizes by choosing tokens that maximize their log-likelihood difference (read the paper if you are interested, not going into more detail here!)

Decoding Hyperparameters & Practical considerations

GENERATING TEXT WITH LLMs

In order to generate text with a generative language model you have to use their built in `generate()` method:

- You need a tokenized input prompt
- This will be the input to the model
- Then you specify the hyperparameters of the models' `generate()` method to control the output length and to choose the desired decoding strategy
- You can find all the code in this notebook: ▸ decoding_examples.ipynb

GREEDY SEARCH

The default decoding strategy is **greedy search**, if all the default hyperparameters are used

Prompt: "Once upon a time"

- `model.generate(tokenized_prompt)`
- **Output:** Once upon a time, the world was a place of great beauty and great danger. The world was a place of great danger, and the world was a place of great danger. The world was a place of great danger, and the world was a place of great danger.

BEAM SEARCH

In order to use **beam search** you simply have to add the `num_beams` argument to the `generate()` method and set it to a value > 1
Prompt: "Once upon a time"

- `model.generate(tokenized_prompt, num_beams=5)`
- Once upon a time, it was said, there would be a time when the world would be a better place. It was a time when the world would be a better place. It was a time when the world would be a better place. It was a time when the world would be a better place. It was a time when the world would be a better place.

SAMPLING WITH TEMPERATURE (1)

In order to activate **sampling** you have to set `do_sample=True`. To additionally use **temperature** you have to set the `temperature`. Also `top_k` has to be set to 0, as 50 is the default value
Prompt: "Once upon a time"

- `model.generate(tokenized_prompt, do_sample=True, temperature=0.7, top_k=0)`
- **Output:** Once upon a time, we could have seen ourselves as a modern day, modern version of ourselves. Ken, who was assembled with great courage and bravery to fight for the cause of women's reproductive rights, has been doing so for more than 15 years, and his career is full of inspiring stories. He is a poet, a civil rights leader, a hard-nosed activist and a true American hero.

SAMPLING WITH TEMPERATURE (2)

If we set `temperature` to a super high value, the output distribution will approximate a uniform distribution. For `temperature` $\rightarrow 0$ the output distribution will have all the probability mass in the most probable token and `generate()` will be equivalent to **greedy search**
Prompt: "Once upon a time"

- `model.generate(tokenized_prompt, do_sample=True, temperature=10000000.0, top_k=0)`
- **Output:** Once upon a time indicators Gran slew extrater rockedIA moaningJohn Chin knocking transgender Button nin forgiving submarine traveller efforts Pascal despairaways Hem Melvin Toryunn Amb velvet

This output is as expected since with such a high `temperature` every token in the vocabulary will have the same probability

TOP-K SAMPLING

In order to activate **top-k sampling** you have to set `do_sample=True` and set `top_k` to a integer > 0. If you use `top_k=1` it will again be equivalent to **greedy search** as only the most probable token is chosen

Prompt: "Once upon a time"

- `model.generate(tokenized_prompt, do_sample=True, top_k=10)`
- **Output:** Once upon a time, he did not think he had a chance and turned to her and said:
  "'My dear, what do you mean by that? I know your father, but he is dead.'
  "So she said:
  'I knew it. I know your father, but I do not know his whereabouts. You know my mother.'

TOP-P SAMPLING

Set `top_p` to some float and `do_sample=True` to use **top-p sampling**. Using a very small value is equal to using **greedy search**. Using `top_p=1.0` will consider all tokens
Prompt: "Once upon a time"

- `model.generate(tokenized_prompt, do_sample=True, top_p=0.7, top_k=0)`
- **Output:** Once upon a time they arrived at the well of water, they saw that their creature was burning in flames. After they had brought their clothes back to the dry land, they were found that they had died in the burning body.

CONTRASTIVE SEARCH

For **contrastive search** you have to set `penalty_alpha` to a float and set `top_k` as previously mentioned. Remember we are **not** sampling here. For `penalty_alpha=0` we only maximize the model confidence term and it becomes **greedy search** again. Using `penalty_alpha=1` we only care about the degeneration penalty term
Prompt: "Once upon a time"

- `model.generate(tokenized_prompt, do_sample=True, penalty_alpha=0.6, top_k=20)`
- **Output:** Once upon a time, the man who had been the object of the most intense and most intense hatred, the most intense and most intense hatred, was the man who had been the object of the most intense and most intense hatred, and the man who had been the object of the most intense and most intense hatred.

CONCLUSION

- In practice you wouldn't use one decoding strategy in isolation
- You would have to play around with the different hyperparameters and decoding strategies until you get a good result
- But how do we determine if a generated text is good or not?
- For that there are several evaluation metrics $\rightarrow$ next chapter

Evaluation Metrics

HOW DO WE EVALUATE LLMs?

How to choose the appropriate evaluation metric?

- Does the task have a gold reference?
  - BLEU score ▸ Papineni et al., 2002
  - ROUGE score ▸ Lin, 2004
- Are we dealing with open ended text generation without a gold reference?
  - Diversity ▸ Su et al., 2022
  - Coherence ▸ Su et al., 2022
  - MAUVE ▸ Pillutla et al., 2021
- If you have the proper resources choose human evaluation

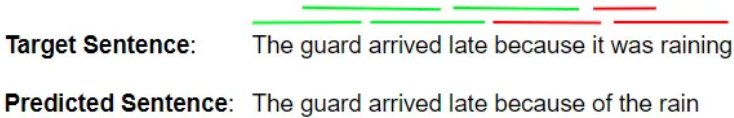
BLEU SCORE (1)

Given a task with a gold reference, e.g machine translation or text summarization, you compare the generated output with the given source reference to compute the BLEU score:

▸ Towards Data Science, Ketan Doshi

Five out of eight 1-grams are correctly predicted:
$\rightarrow p_1 = 5/8$

BLEU SCORE (2)



▸ Towards Data Science, Ketan Doshi

Four out of seven 2-grams are correctly predicted:
$\rightarrow p_2 = 4/7$
You keep doing this procedure until N n-grams and compute a weighted geometric average over the precision scores with weights w_n:

$$exp\left(\sum_{n=1}^{N} w_n \cdot log(p_n)\right)$$

BLEU SCORE - BREVITY PENALTY

In order to penalize very short predictions (it's more likely for shorter sentences to achieve a good precision score) the BLEU score additionally has a brevity penalty term:

$$BP = \begin{cases} 1, & \text{if } c > r \\ e^{(1-r/c)}, & \text{if } c \leq r \end{cases}$$

- With r being the **reference corpus length** and c the **candidate corpus length**
- The final formula is then:

$$BLEU = BP \cdot exp\left(\sum_{n=1}^{N} w_n \cdot log(p_n)\right)$$

ROUGE SCORE

- The ROUGE (Recall-Oriented Understudy for Gisting Evaluation) is a metric commonly used for evaluating the quality of machine-generated text, particularly summaries
- ROUGE measures the similarity between the generated summary and one or more reference (human-written) summaries
- ROUGE includes multiple metrics, such as ROUGE-N (for n-grams), ROUGE-L (for longest common subsequence), and ROUGE-W (for weighted n-grams). Depending on the task, these metrics capture different aspects of summary quality, allowing a more comprehensive evaluation

EXAMPLE: ROUGE-1 PRECISION

Consider the following source sentence S and candidate summary C:

- **S:** The cat is on the mat.
- **C:** The cat and the dog.

Using the ROUGE-N precision score with $N = 1$ you get:

- Three correctly predicted unigrams
- Total of number of unigrams in C is 5

$\rightarrow$ ROUGE-1 precision $= 3/5 = 0.6$

There are more ROUGE scores as mentioned earlier. You can find more details here: ▸ Medium, Fabio Chiusano

METRICS WITHOUT A GOLD REFERENCE

- BLEU and ROUGE are both used for tasks that have a gold reference you can compare your prediction to
- In open ended text generation you just have a prompt and an output generated by the model
- You don't have any gold reference to compare your output to
- Therefore you have to get a bit more creative with the choice of evaluation metrics

DIVERSITY

Su et al., 2022 *define diversity in their paper, where they introduce contrastive search, as the generation repetition at different n-gram levels*:

- **Generation Repitition:**
  - Measures the sequence-level repitition as the portion of duplicate *n*-grams in the generated text
  - For a generated text continuation x_{cont} the repitition at *n*-gram level is defined as:

$$\text{rep-n} = 100 \times \left(1.0 - \frac{|\text{unique } n\text{-grams}(x_{cont})|}{|\text{total } n\text{-grams}(x_{cont})|}\right)$$

DIVERSITY (2)

- **Diversity:**
  - Repitition at different n-gram levels:

$$\text{DIV} = \prod_{n=2}^{4} \left(1.0 - \frac{\text{rep-n}}{100}\right)$$

  - Plugging in rep-n from the previous slide, this expression simplifies to:

$$\text{DIV} = \prod_{n=2}^{4} \frac{|\text{unique } n\text{-grams}(x_{cont})|}{|\text{total } n\text{-grams}(x_{cont})|}$$

  - A low diversity score suggests the model suffers from repitition, and a high diversity score means the model-generated text is lexically diverse

COHERENCE

This measure was also proposed by Su et al., 2022 *as the cosine similarity between the sentence embeddings of the prompt* x_{prompt} *and a generated text conitinuation* x_{cont}:

- They use pre-trained SimCSE sentence embeddings $EMB(x)$ proposed by Gao et al., 2022:

$$COH(x_{cont}, x_{prompt}) = \frac{EMB(x_{prompt}) \cdot EMB(x_{cont})}{\|EMB(x_{prompt})\| \cdot \|EMB(x_{cont})\|}$$

- The higher the coherence-score the better the model-generated text fits to the given prompt

MAUVE

▸ Pillutla et al., 2021

- A language model is an estimate $\hat{P}(x)$ of the probability distribution over sequences of text $x = (x_1, ..., x_{|x|})$, consisting of tokens x_t belonging to a fixed vocabulary
- Given a context $x_{1:t}$, a language model $\hat{P}$ and a stochastic decoding strategy we generate text by sampling $\hat{x}_{t+1} \sim \hat{P}(\cdot|x_{1:t}), \hat{x}_{t+2} \sim \hat{P}(\cdot|x_{1:t}, \hat{x}_{t+1})$, etc.
- The decoding strategy and the language model taken together define a distribution Q over text, which we call *model distribution*
- The goal of MAUVE is to measure the gap between the model distribution Q and the target distribution P

SOURCES OF ERROR IN TEXT GENERATION

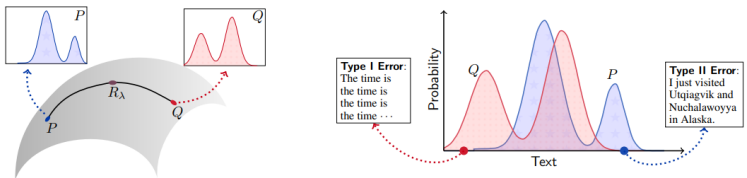


Figure 1: **Left**: MAUVE compares the machine text distribution Q to that of human text P by using the family of mixtures $R_\lambda = \lambda P + (1-\lambda)Q$ for $\lambda \in (0, 1)$. **Right**: Illustration of *Type I errors*, where Q produces degenerate, repetitive text which is unlikely under P, and, *Type II errors*, where Q cannot produce plausible human text due to truncation heuristics [26]. MAUVE measures these errors softly, by using the mixture distribution R_λ. Varying λ in $(0, 1)$ gives a divergence curve and captures a spectrum of soft Type I and Type II errors. MAUVE summarizes the entire divergence curve in a single scalar as the area under this curve.

- The gap between Q and P arises from two sources of error

SOURCES OF ERROR IN TEXT GENERATION / 2

- Type I error: Q places high mass on text which is unlikely under P
- Type II error: Q cannot generate text which is plausible under P
- They formalize the two errors through the Kullback-Leibler divergence:
  - $KL(Q|P)$ penalizes Q if there is a text x that leads to a high $Q(x)$ but a low $P(x)$, which is the Type I error
  - Similarly the Type II error is defined by $KL(P|Q)$
- Issue: both KL divergences are infinite if the supports of Q and P are not identical

SOURCES OF ERROR IN TEXT GENERATION / 3

- The authors overcome this issue by *softly* measuring the two errors with a mixture distribution:

$$R_\lambda = \lambda P(1-\lambda)Q \quad \text{for} \quad \lambda \in (0,1)$$

- (soft) Type I error: $KL(Q, R_\lambda)$
- (soft) Type II error: $KL(P, R_\lambda)$

DIVERGENCE CURVE

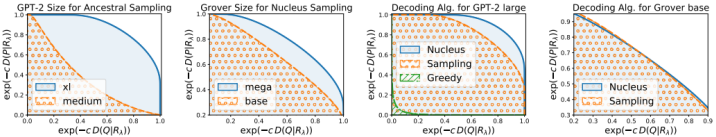


Figure 2: Divergence curves for different models (GPT-2 [45], Grover [61]) and decoding algorithms (greedy decoding, ancestral and nucleus sampling). MAUVE is computed as the area of the shaded region, and larger values of MAUVE indicate that Q is closer to P. In general, MAUVE indicates that generations from larger models and nucleus sampling are closer to human text. **Rightmost**: Nucleus sampling has a slightly smaller Type I error than ancestral sampling but a higher Type II error, indicating that ancestral sampling with Grover base produces more degenerate text while nucleus sampling does not effectively cover the human text distribution.

- To capture all the possible values of the mixture weight λ they vary λ between 0 and 1 to generate a *divergence curve*:

$$C(P,Q) = \{(exp(-cKL(Q|R_\lambda)), exp(-cKL(P|R_\lambda)) : R_\lambda = \lambda P + (1-\lambda)Q, \lambda \in (0,1)\}$$

DIVERGENCE CURVE / 2

- $MAUVE(P, Q)$ is the area under this divergence curve, it is a summary of the trade-off between Type I and II errors and lies in (0,1] (more details can be found in the paper ▸ Pillutla et al., 2021)

HUMAN EVALUATION

Why Human Evaluation?

- **Subjectivity of Quality**: Human judgments are essential for evaluating the nuanced quality of text that automatic metrics might miss, such as humor, creativity, and relevance

Key Considerations

- **Evaluators**: Use domain experts or crowdworkers, depending on the task complexity
- **Evaluation Criteria**:
  - **Fluency**: Is the generated text grammatically correct and natural-sounding?
  - **Coherence**: Does the text make logical sense?
  - **Diversity**: Is the output lexically diverse?
- **Challenges**:

HUMAN EVALUATION / 2

- **Subjectivity**: Different evaluators might have varying opinions, leading to inconsistency
- **Cost and Time**: Human evaluation is resource-intensive
- **Bias**: Evaluators might bring in their biases, which can skew results

Fine-Tuning and Prompting

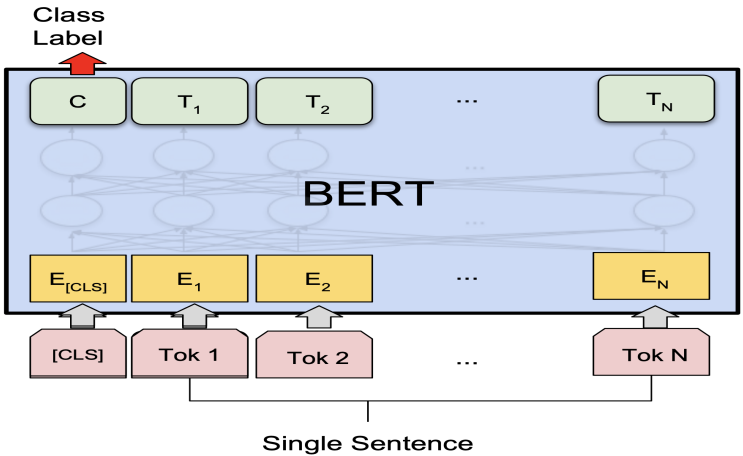
RECAP

- language modeling objective
  - masking and next token prediction
  - no explicit understanding of tasks
- this is true for encoder and decoder models
- So we need to do additional work if we want to use language models for solving tasks!

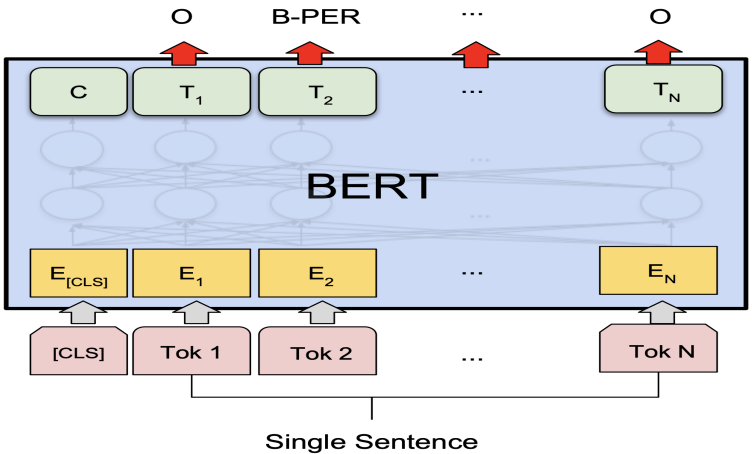
HOW TO SOLVE A TASK WITH LANGUAGE MODELS? (1)

- "old-style" single-task fine-tuning
  - supervised training of the pretrained model on a task-specific training set of size k (where k is not small, e.g., $k = 100$)

BERT FINETUNING EXAMPLE 1



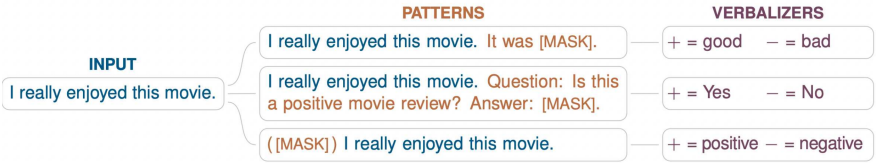
BERT FINETUNING EXAMPLE 2



HOW TO SOLVE A TASK WITH LANGUAGE MODELS? (1)

- "old-style" single-task fine-tuning
  - supervised training of the pretrained model on a task-specific training set of size k (where k is not small, e.g., $k = 100$)
  - the output can be an arbitrary category (e.g., "0", "1" for sentiment analysis)
  - (this is the typical way encoder models like BERT are used)
  - alternatively, the output can be a meaningful "verbalizer" (e.g., "negative", "positive" for sentiment analysis)
    ▸ Schick et al., 2020
  - still very much relevant if you need to deploy a small efficient model for a focused task

VERBALIZER


PATTERNS
VERBALIZERS
INPUT
I really enjoyed this movie.
I really enjoyed this movie. It was [MASK].
+ = good − = bad
I really enjoyed this movie. Question: Is this a positive movie review? Answer: [MASK].
+ = Yes − = No
([MASK]) I really enjoyed this movie.
+ = positive − = negative

VERBALIZATION: SUMMARY

- Finetuning a language model on predicting meaningful labels ("good", "bad") requires less data than finetuning on classical NLP labels like "0" and "1".
- Reason: The language model understands the labels already, so it doesn't have to learn them.

HOW TO SOLVE A TASK WITH LANGUAGE MODELS? (2)

- few-shot prompting
  - provide, in-context, k (where k is small, e.g., $k = 5$) examples of what the model is supposed to do
  - the model will then often complete the task just based on analogy to these few shots
  - this is a typical way of using current autoregressive models for tasks
  - no change to the parameters of the model, i.e., no training

FEW-SHOT PROMPTING EXAMPLE

```
1  Translate English to French:
2  sea otter => loutre de mer
3  peppermint => menthe poivrée
4  plush girafe => girafe peluche
5  cheese =>
```

(from gpt3 paper – "Translate English to French" should appear four times to make it a 3-shot example that has 3 identical shots as required by the definition of few-shot prompting)

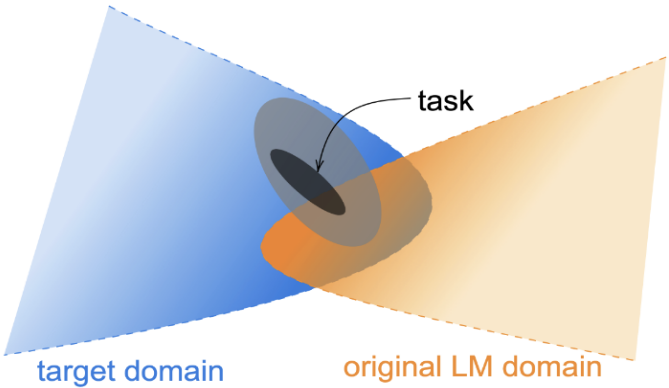
HOW TO SOLVE A TASK WITH LANGUAGE MODELS? (3)

- multi-task finetuning
  - finetuning on a large training set of *many* tasks
  - the more tasks the better
  - method 1: consistent task format, e.g., each task is reformulated into a question-answering format
  - method 2: more open task format
  - ideally: task format includes "instructions", similar to verbalizer
  - Examples: T0, T5 and FLAN, see below

OTHER TYPES OF FINETUNING (1)

- Finetuning has yet another meaning.
- Continued pretraining is also sometimes called finetuning.
- Continued pretraining: given a language model that has been trained on generic data (web, reddit etc), adapt it to a new domain (e.g., company-internal data) by training it on a large corpus from this new domain.
- objective: standard language modeling objective
- This results in a language model that has all the nice capabilities of a generic language model, but also understands the special domain.
- Not trivial to do well

CONTINUED PRETRAINING



TERMINOLOGY

- This lecture
  - single-task finetuning
  - multi-task finetuning
  - prompting
    (this is definitely not finetuning)
  - continued pretraining
  - **Question:** terminology clear?
- Next lecture
  - instruction tuning
    (can be seen as a form of finetuning)
  - reinforcement learning
    (can also be done through a form of finetuning)

ISSUES WITH SINGLE-TASK FINE-TUNING

- The result is a single-task model.
  - Generalization of the model is only w.r.t. to one task / data distribution
- Requires annotated data (e.g., $k = 100$)
- **Question:** Could there be other tasks that might benefit?

ISSUES WITH FEW-SHOT PROMPTING: (1) CLASSIFICATION

- Assumption: Model has learned about the task during (unsupervised) pre-training
- Write prompt so that a direct response must be given by the language model.
  - Just a label, just yes/no answer, just a name in QA
- This is not natural dialogic behavior: humans typically don't just answer with a label, yes/no, a name (although sometimes they do)
- See next lecture
- Current use of the word "prompt" is more general: basically anything you type into the LLM context window

ISSUES WITH FEW-SHOT PROMPTING: (2) TEXT GENERATION

- Again, assumption: Model has learned about the task during (unsupervised) pre-training
- Again: Write prompt so that a direct response must be given by the language model.
- Works best if the task has occurred during unsupervised training.
- **Question:** For which tasks is this expected to work well?

MULTI-TASK LEARNING: FLAN/T0/T5

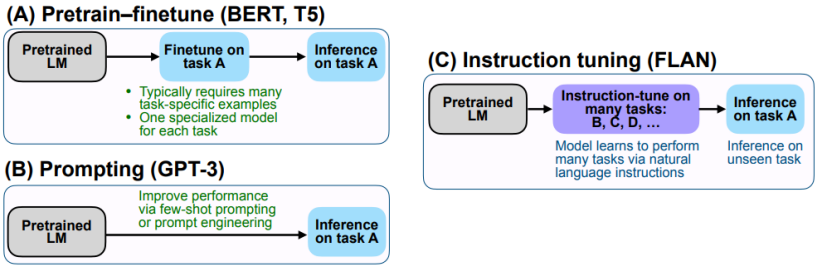
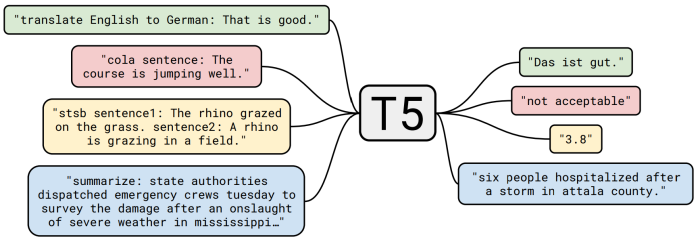


Figure 2: Comparing instruction tuning with pretrain–finetune and prompting.

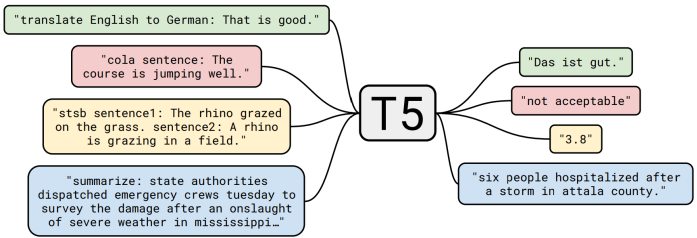
▸ Source: Wei et al., 2021

INITIAL APPROACH: MULTITASK FINETUNING: BEST OF BOTH WORLDS (FINETUNING AND PROMPTING)



▸ Source: Raffel et al., 2020

INITIAL APPROACH: MULTITASK FINETUNING: BEST OF BOTH WORLDS (FINETUNING AND PROMPTING)



▸ Source: Raffel et al., 2020

Question: What exactly does the model learn here? How well would we expect this to generalize to new tasks?

MULTITASK PROMPTED TRAINING

- Multitask Prompted Training involves learning from multiple tasks using **unified prompt formats** as a means to improve generalization to new, unseen tasks.
- (as opposed to T5: non-unified)
- This means that the model can perform well on tasks it hasn't been explicitly trained on.
- The key for this lies in the set of shared prompts it has learned from during fine-tuning.

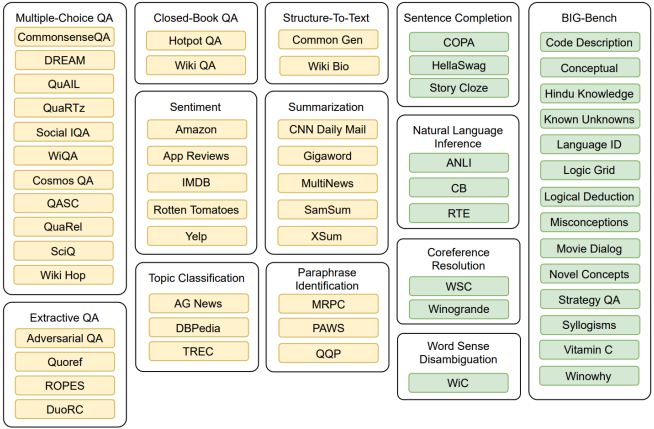
MULTITASK PROMPTED TRAINING

- Benchmark for Evaluation:
  - Instead of using held-out *samples* (as is standard in NLP) ...
  - ... we now use held-out *tasks*
  - All data sets belonging to a held-out task go to the test set
  - Generalization across tasks
- The "prompts" are key here to facilitate transfer to new tasks: the model can solve new tasks by generalizing from the train prompts and the ability to generate text outputs.

TYPICAL NLP TASK: COREFERENCE RESOLUTION

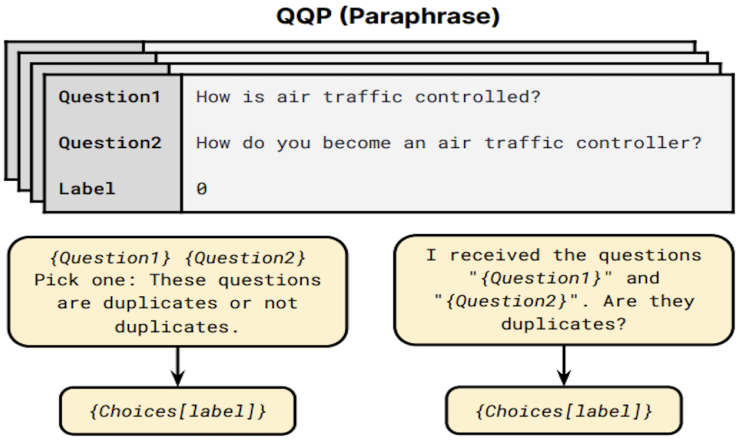
- One of the tasks on the next slide
- In the simplest case the task is to determine which noun phrase a referring expression like "she" or "this person" is referring to.
- Example: "Peter helped Paul because he was familiar with the problem."
- What does "he" refer to?

T0: TRAINING AND TEST TASKS



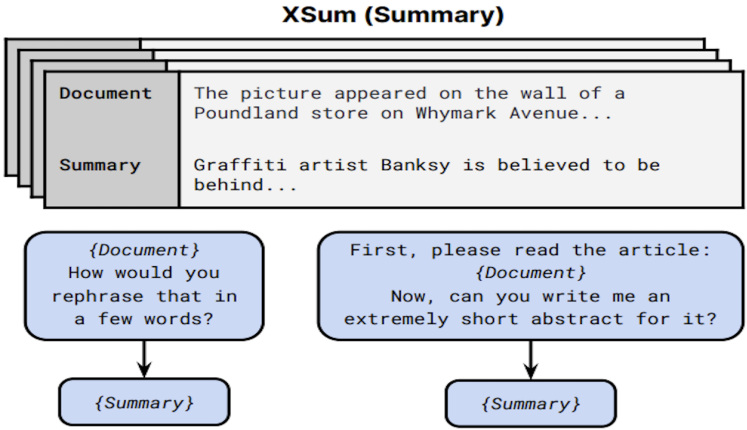
▸ Source: Sanh et al., 2021 **Question:** What is an easy task? What is a hard task?

T0 – PROMPT TEMPLATES



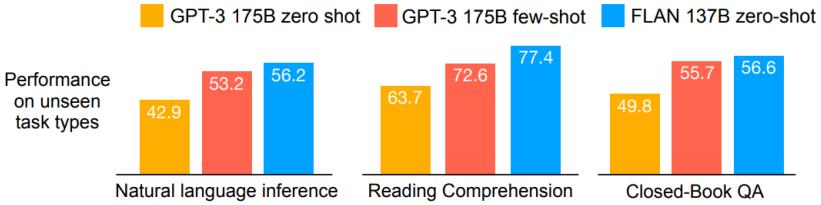
▸ Source: Sanh et al., 2021

T0 – PROMPT TEMPLATES



▸ Source: Sanh et al., 2021

EXAMPLE OF EVALUATION: FLAN



▸ Source: Wei et al., 2021

MULTI-TASK TRAINING VS INSTRUCTION TUNING

- The term "instruction tuning" was introduced by FLAN paper,
  ▸ Source: Wei et al., 2021
- However, today "instruction tuning" refers to a wide variety of methods that train on instruction-output pairs, not just tasks
- In particular, open-ended generation and "alignment" is now included under the rubric "instruction tuning": brainstorming, writing a joke, give me ten analogies for concept X, don't use hate language, don't take positions on controversial political issues etc
- These are not classical NLP tasks.
- See next lecture

MULTI-TASK TRAINING VS INSTRUCTION TUNING / 2

- Our use of terminology in this class:
- multi-task finetuning = tasks in the classical NLP sense
- instruction-tuning: a broad approach to changing the behavior of a raw language model (i.e., trained on next-word prediction) to be "helpful, harmless and honest". This includes training on classical NLP tasks, but it's just one part.

FINE-TUNING CONCLUSIONS

- The more training tasks the better
- Multi-task finetuning generalizes across models
  - It works well on different architectures
- Greatly improves usability
- It is relatively compute-efficient
  - Pretraining is hugely expensive, finetuning and instruction tuning are usually much cheaper.
  - For PaLM 540 B it takes 0.2 % of pre-training compute, but improves by 9.4 %

SLIDO

- #1312409 slido

PROMPTING: DEFINITION

- Prompting always means that you do not finetune the model, i.e., you don't change the parameters of the model.
- The term prompting was initially used for when you carefully design the input to the language model, so that you will get the desired output.
- But nowadays any type of input to the model can be called a prompt.
- Few-shot or k-shot prompting means that you give a few examples or k examples of what you want the model to do as part of the input to the model.
- Zero-shot prompting means that you do not give an example. The term zero-shot prompting is often used in the context of few-shot prompting (contrasting using $k > 0$ examples vs $k = 0$ examples).

Chain-of-Thought Few-Shot Prompting

Chain-of-thought few-shot prompting MOTIVATION

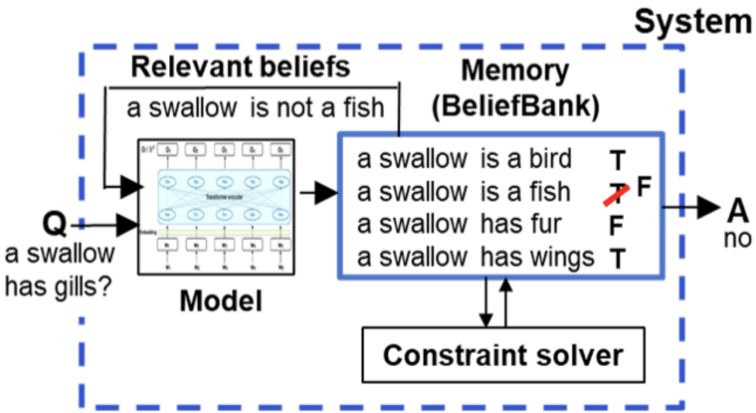
How to boost the reasoning capabilities of LMs? ▸ Wei et al., 2021

- Use formal approaches, e.g., logic, symbolic reasoning
  - Example: ▸ BeliefBank
  - Difficult to train and deploy, not widely used
- Standard few-shot learning via prompting works for many tasks
  - Still, it works poorly for many tasks that require reasoning
- COT-FS-P
  - A form of few-shot prompting
  - Each "training example" has the form:
    <input, *chain of thought*, output>
  - chain of thought:
    series of reasoning steps that lead to the final answer

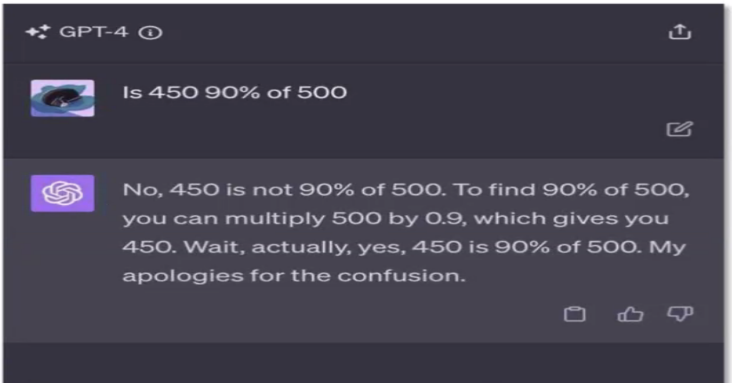
Chain-of-thought few-shot prompting MOTIVATION / 2

- application: reasoning tasks: complex, commonsense, symbolic etc

NEUROSYMBOLIC APPROACH (CURRENTLY INFREQUENTLY USED)

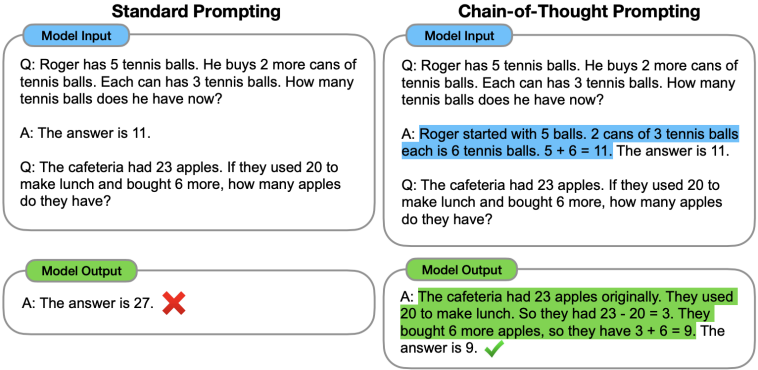


LMS NOT GOOD AT REASONING TASKS (THIS IS WHY WE NEED COT-FS-P.)



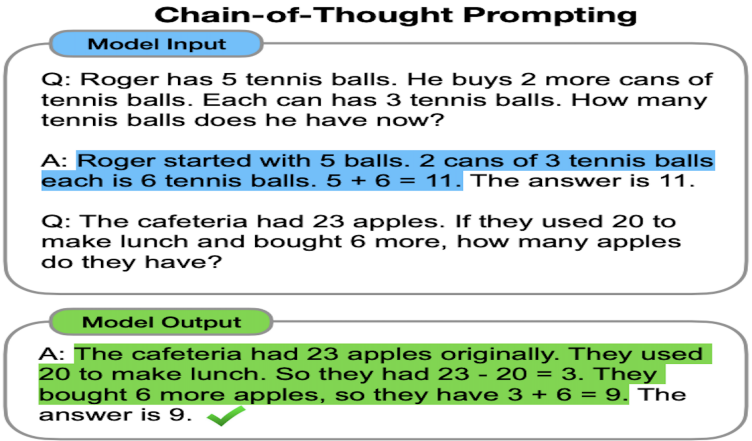
Question: What is the problem here?

Chain-of-thought few-shot prompting PARADIGM



▸ Source: Wei et al., 2022

chain-of-thought few-shot prompting PARADIGM



BENEFITS OF chain-of-thought few-shot prompting

- Decompose multi-step problems and thus allocate more compute to problems requiring more reasoning steps.
- This helps LMs solve problems they otherwise would not be able to solve.
- By describing the reasoning, interpretability is increased. It provides the possibility to observe where reasoning went wrong
- COT-FS-P is closer to how humans solve tasks using language

chain-of-thought few-shot prompting

Following slides: Examples of <input, chain of thought, output> triples for commonsense and symbolic reasoning

▸ Source: Wei et al., 2022

The few shots (training examples) are omitted from these examples to save space, but this is chain-of-thought few-shot prompting, that is, the model is prompted with examples of chain of thought reasoning.

EXAMPLES

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3, which is less than water. Thus, a pear would float. So the answer is no.

EXAMPLES

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

EXAMPLES

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

EXAMPLES

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.
Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

EXAMPLES

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.

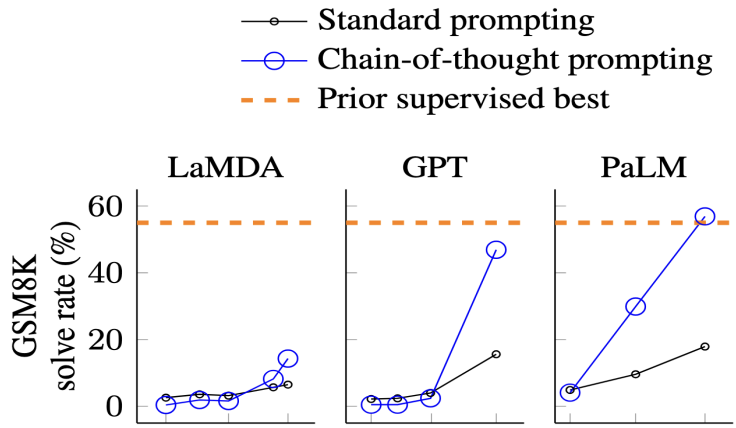
EXAMPLES

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

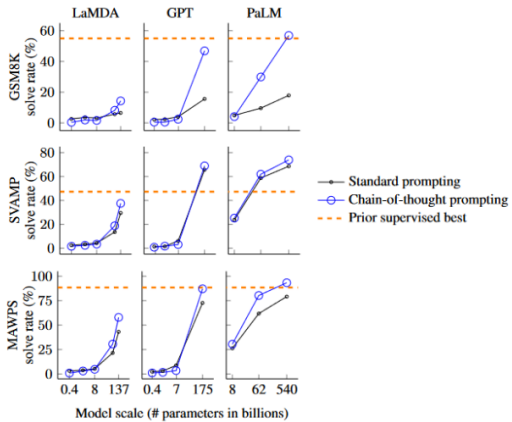
COT-FS-P IMPROVES ARITHMETIC



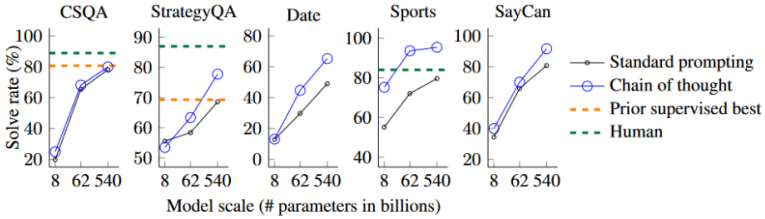
COT-FS-P IMPROVES ARITHMETIC

SVAMP: math word problems with varying structures; MAWPS: repository unifying math problems from different sources;

▸ Source: Wei et al., 2022



COT-FS-P IMPROVES COMMONSENSE



▸ Source: Wei et al., 2022

Emergent Abilities

EMERGENT ABILITIES

An *emergent ability* is an ability that is not present in small models but is present in large models.

- Assumption: Everything else is held constant.
- Questions: Is emergence a rare phenomenon? Are many tasks emergent?
- We can answer this by investigating model families with many sizes.
  - GPT-3
  - Chinchilla
  - PaLM

EMERGENT ABILITIES AND MODEL SIZE

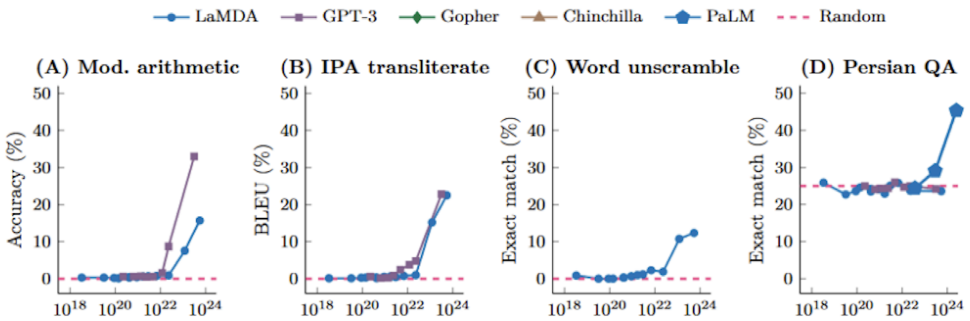


Figure: Examples of emergence in few-shot prompting. ▸ Wei et al., 2022

EMERGENT ABILITIES AND MODEL SIZE

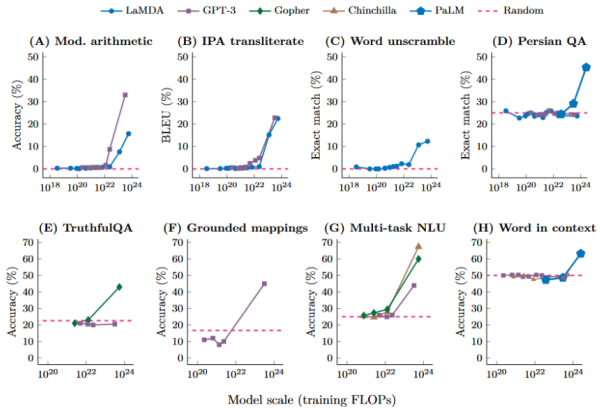


Figure: Examples of emergence in few-shot prompting. ▸ Wei et al., 2022

EMERGENT TASKS IN BIG-BENCH

- "MODEL SIZE (TASK)" means: MODEL can do TASK with SIZE, but not with less than SIZE (hence emerging)
- GPT-3 13B (2 tasks): hindu knowledge, modified arithmetic
- GPT-3 175B (15 tasks): analytic entailment, codenames, phrase relatedness, question answer creation, self evaluation tutoring, ...
- LaMDA 137B (8 tasks): gender inclusive sentences german, repeat copy logic, sports understanding, ...
- PaLM 8B (3 tasks): auto debugging, sufficient information, ParsiNLU reading comprehension
- PaLM 64B (14 tasks): anachronisms, ascii word recognition, conceptual combinations, ...
- PaLM 540B (25 tasks): analogical similarity, causal judgment, code line description, crass ai, cs algorithms, ...

EMERGENT TASKS IN MMLU

- Chinchilla 7B (7 tasks): Professional Medicine, High School Statistics, High School Macroeconomics, High School Psychology, Anatomy, High School Government And Politics, High School Microeconomics
- Chinchilla 70B (44 tasks): International Law, Human Aging, Sociology, Us Foreign Policy, High School World History, Marketing, Logical Fallacies, Miscellaneous, College Biology, High School Us History, Security Studies, High School European History, ...

OTHER EMERGENT TASKS

- GPT-3 paper: 3 digit addition/subtraction (GPT-3 13B), 4-5 digit addition/substraction (GPT-3 175B), leveraging few-shot examples for word denoising (GPT-3 13B)
- Gopher paper: Toxicity classification (Gopher 7.1B), TruthfulQA (Gopher 280B)
- Patel & Pavlick: grounded conceptual mappings (GPT-3 175B)
- PaLM paper: Word in Context benchmark (PaLM 540B)

COUNTER ARGUMENT

▸ Source: Schaeffer et al., 2023

Two defining properties for emergent abilities in LLMs:

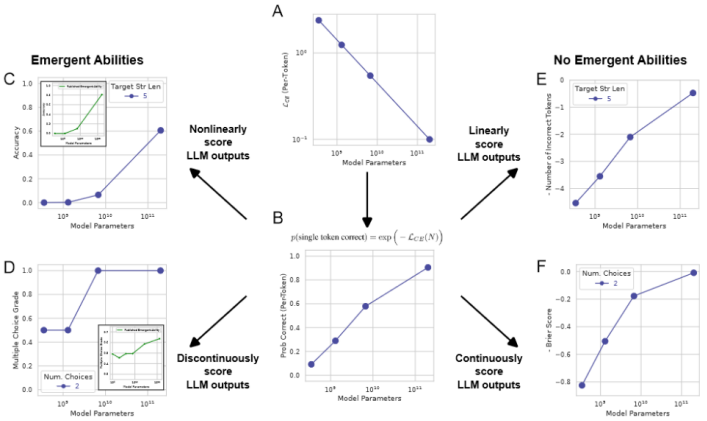
1. **Sharpness:** transitioning seemingly instantaneously from not present to present
2. **Unpredictability:** transitioning at seemingly unforeseeable model scales

Claim: Emergent abilities appear only under metrics that nonlinearly or discontinuously scale model's per-token error rate:

$$\text{Multiple Choice Grade} \overset{\text{def}}{=} \begin{cases} 1 & \text{if highest probability mass on correct option} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Exact String Match} \overset{\text{def}}{=} \begin{cases} 1 & \text{if output string exactly matches target string} \\ 0 & \text{otherwise} \end{cases}$$

COUNTER ARGUMENT



▸ Source: Schaeffer et al., 2023

COUNTERARGUMENT: SUMMARY

- Emergence is merely an artifact of the evaluation measures we use.
- If TASK is emergent for family MODEL at SIZE on METRIC, then it is often possible to choose another metric for which the task is not emergent.
- Emergent abilities can be induced in computer vision tasks as well
- A task and a metric are distinct and meaningful choices when constructing a benchmark
- When choosing a metric, one should consider the metric's effect on the per-token error rate; may require adapting the measuring process

COUNTER COUNTER ARGUMENT (1)

https://www.journals.uchicago.edu/toc/an/current, Robert Dryja

Emergence is a holistic / gestalt phenomenon. Even if there is a straightforward mathematical explanation for an emergent phenomenon, it can still make sense to distinguish it from subjectively non-emergent phenomena.

COUNTER COUNTER ARGUMENT (2)

- **Question:** Have you had a personal emergence experience with AI or do you think you will have one in the future?

BACK TO FINETUNING

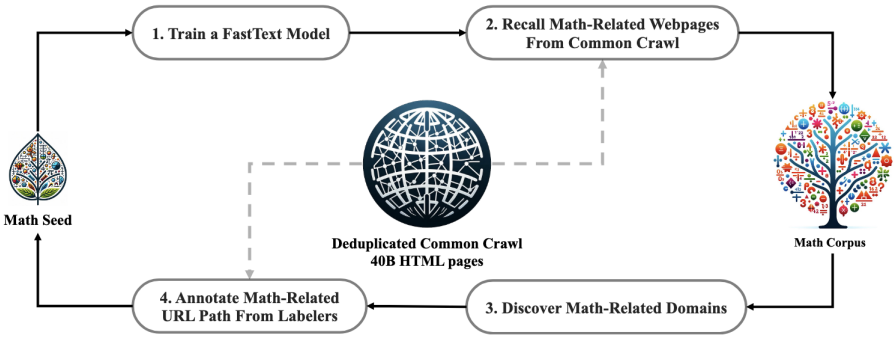
- **Question:** You are working at a telecommunications company. You are given the task of converting a German manual from polite form ("Wir sind jederzeit unter der folgenden Nummer fuer Sie erreichbar") to familiar form ("Wir sind jederzeit unter der folgenden Nummer fuer Dich erreichbar"). How would you do this?

DeepSeek Reasoning

SOURCES

- First major open reasoning LM (as opposed to proprietary closed): DeepSeek
- ▸ DeepSeek-AI, 2025
- ▸ Nature, 2025

DATA FOR SFT: 40B TOKENS!



▸ DeepSeekMath, 2024

REINFORCEMENT LEARNING

- In reinforcement learning, an agent takes a sequence of actions towards a goal and gets feedback on the success of that sequence at the end.
- Details: Next lecture
- For reasoning trajectories, the feedback is: correct or incorrect.
- (thanks to Emma Brunskill)

	supervised learning	reinforcement learning
goal	optimal predictor	optimal policy
learn from experience?	yes	yes
generalization?	yes	yes
feedback	rich (each token)	sparse (in/correct)
feedback when?	immediately	delayed
exploration?	no	yes

REASONING TRACES: FORMAT

- The User asks a question and the Assistant solves it.
- The Assistant first thinks about the reasoning process "in its mind" and then provides the answer to the User.
- The reasoning process and answer are enclosed within <think>...</think> and <answer>...</answer> tags
- `<think> reasoning process here </think>`
- `<answer> answer here </answer>`
- "We intentionally limit our constraints to this structural format, avoiding any content-specific biases to ensure that we can accurately observe the natural progression of the model during the RL process."
- So this is not a few-shot approach!

EXAMPLE OF A REASONING TRACE

Query

The denominator of a fraction is 7 less than 3 times the numerator. If the fraction is equivalent to 2/5, what is the numerator?

Correct reasoning trace

Step 1	*Let the numerator be x.*
Step 2	*The denominator is 3x-7.*
Step 3	*We know that x/(3x-7) = 2/5.*
Step 4	*Therefore, 5x = 6x-14.*
Step 5	*Finally, we get x= 14.* (Correct)

Question:
Is this chain of thought few-shot prompting?

AUTOMATIC, NON-HUMAN REWARDS FOR LM-GENERATED REASONING TRACES

Accuracy rewards evaluate whether the response is correct. For example, in the case of maths problems with deterministic results, the model is required to provide the final answer in a specified format (for example, within a box), enabling reliable rule-based verification of correctness. Similarly, for code competition prompts, a compiler can be used to evaluate the responses of the model against a suite of predefined test cases, thereby generating objective feedback on correctness.

AUTOMATIC VERIFIER/VERIFIABLE REWARDS: AIME EXAMPLE

PROMPT

Let $b \geq 2$ be an integer. Call a positive integer n b-*eautiful* if it has exactly two digits when expressed in base b, and these two digits sum to $\sqrt{n}$. For example, 81 is 13-eautiful because $81 = \underline{6}\ \underline{3}_{13}$ and $6 + 3 = \sqrt{81}$. Find the least integer $b \geq 2$ for which there are more than ten b-eautiful integers.

Please reason step by step, and put your final answer within \boxed{}.

Evaluation

Parse the final answer within \boxed{} and use a rule-based grader to determine if it equals the ground truth. Round numerical values as needed, and use 'SymPy' to parse expressions.

INFERENCE TIME SCALING

- For a hard problem, it often helps to think for a long time to come up with the solution.
- Inference time scaling: teach the model to think for a while before giving the answer.
- Here, thinking means that the model produces potentially long output, the reasoning trace, that leads to the correct answer.

WHAT ELEMENTS OF REASONING ARE LEARNED?

- dynamic strategy adaptation
- self-verification
- reflection
- backtracking
- reasoning traces are getting longer
- exploration of alternative approches
- diverse and sophisticated reasoning behaviors

AHA MOMENT

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

$\cdots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

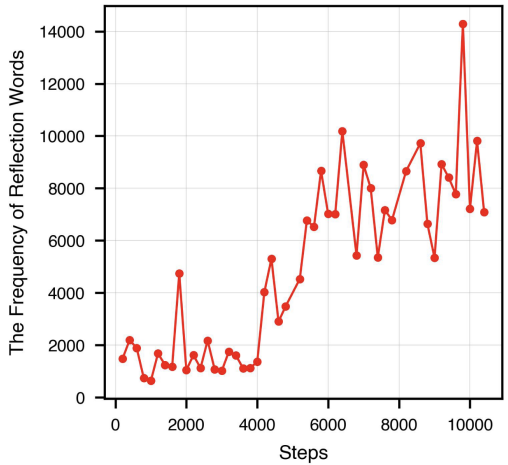
ANOTHER EXAMPLE FOR AHA MOMENT

How many r in raspberry? **Question**

Let's break down the process of counting the letter 'r' in the word "raspberry" ...
* First letter: 'r' - This is an 'r', count = 1.
* Second letter: 'a' - Not an 'r', count remains 1 ...
* Sixth letter: 'e' - Not an 'r', count remains 1.
* Seventh letter: 'r' - This is an 'r', count = 2.
* Eighth letter: 'y' - Not an 'r', count remains 2 ...
The number of 'r's in "raspberry" is 2.
Wait, let's re-read the question carefully. It asks "How many r in raspberry?" ... * r - a - s - p - b - e - r - r - y ... * First 'r' ... * Second 'r' ... * Third 'r' ... Count = 3 ... **Reasoning trace**

My initial answer of 2 was incorrect due to a quick reading of the word. **Final Answer:** The final answer is $\boxed{3}$ **Response**

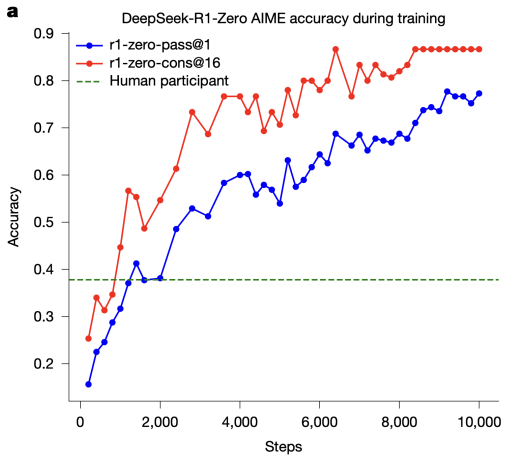
THE RISE OF REFLECTIVE TERMS



Frequency of reflective terms (‘wait’, ‘mistake’, ‘however’, ‘but’, ‘retry’, ‘error’, ‘verify’, ‘wrong’, ‘evaluate’, ‘check’) in model output during training

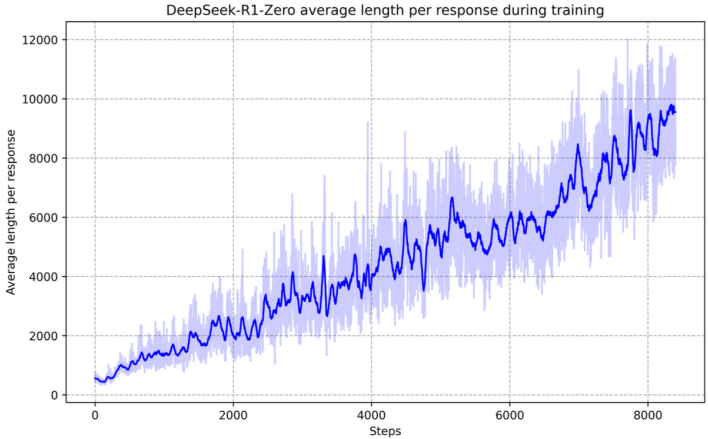
ACCURACY AIME: ABOVE HUMAN AVERAGE

ACCURACY AIME: ABOVE HUMAN AVERAGE / 2



R1 SCALING: WITHOUT EXPLICIT LENGTH SIGNAL

R1 SCALING: WITHOUT EXPLICIT LENGTH SIGNAL / 2



UNFAITHFULNESS OF LMS TO TRUE REASONING

```
Human: Answer in one word. What is 36+59?

Assistant: 95

Human: Briefly, how did you get that?

Assistant: I added the ones (6+9=15), carried the 1, then added the tens
(3+5+1=9), resulting in 95.
```

UNFAITHFULNESS OF LMS TO TRUE REASONING

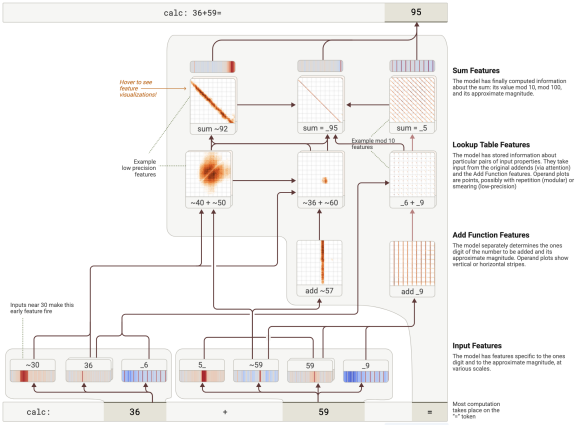


Figure 26: A simplified attribution graph of Haiku adding two-digit numbers. Features of the inputs feed into separable processing pathways.

THINGS WE HAVE NOT COVERED

- use an eco system of several models:
  for distillation, generation, verification
- some "light" human post-editing of training data
- rejection sampling:
  keep only high-quality, high-confidence training data
- apart from math, they also do this for code and other domains
- major hardware optimization work
- claim: cost of training $<$ 1 000 000 million dollar

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